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FOR THE EASTERN DISTRICT OF VIRGINIA
Alexandria Division

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STATE OF RHODE ISLAND
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and

STATE OF TENNESSEE
P.O. Box 20207
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Plaintiffs,

v.

GOOGLE LLC
1600 Amphitheatre Parkway
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Defendant.

COMPLAINT

Table of Contents

I.	Introduction.....	1
II.	Nature of this Action.....	4
III.	Display Advertising Transactions.....	16
A.	How Ad Tech Tools Work.....	16
B.	How Ad Tech Intermediaries Get Paid.....	22
C.	How Publishers and Advertisers Select Ad Tech Tools	24
D.	Why Scale and the Resulting Network Effects are Necessary to Compete in Ad Tech.....	26
E.	How Multi-Homing Enables Competition in the Ad Tech Stack.....	28
IV.	Google’s Scheme to Dominate the Ad Tech Stack.....	30
A.	Google Buys Control of the Key Tools that Link Publishers and Advertisers.....	31
B.	Google Uses Its Acquisitions and Position Across the Ad Tech Stack to Lock Out Rivals and Control Each Key Ad Tech Tool	35
1.	Google Thwarts Fair Competition by Making Its Google Ads’ Advertiser Demand Exclusive to Its Own Ad Exchange, AdX.....	37
2.	In Turn, Google Makes Its Ad Exchange’s Real-Time Bids Exclusive to Its Publisher Ad Server	43
3.	Finally, Google Uses Its Control of Publisher Inventory to Force More Valuable Transactions Through Its Ad Exchange	46
4.	Google’s Dominance Across the Ad Tech Stack Gives It the Unique Ability to Manipulate Auctions to Protect Its Position, Hinder Rivals, and Work Against Its Own Customers’ Interests	55
a)	Google Works Against the Interests of Its Google Ads’ Customers By Submitting Two Bids Into AdX Auctions.....	57
b)	Google Manipulates Its Fees to Keep More High-Value Impressions Out of the Hands of Rivals	60
C.	Google Buys and Kills a Burgeoning Competitor and Then Tightens the Screws.....	65
1.	Google Extinguishes AdMeld’s Potential Threat	65
2.	Google Doubles Down on Preventing Rival Publisher Ad Servers from Accessing AdX and Google Ads’ Demand	68
3.	Google Manipulates Google Ads’ Bidding Strategy to Block Publisher Partnerships with Rivals	71
D.	Google Responds to the Threat of Header Bidding by Further Excluding Rivals and Reinforcing Its Dominance	72

1.	The Industry Attempts to Rebel Against Google’s Exclusionary Practices	72
2.	Google Blunts Header Bidding By “Drying Out” the Competition	78
a)	Google Develops So-Called Open Bidding, Its Own Google-Friendly Version of Header Bidding To Preserve Its Control Over the Sale of Publisher Inventory.....	78
b)	Google Further Stunts Header Bidding by Working to Bring Facebook and Amazon into Its Open Bidding Fold.....	82
c)	Google Manipulates Its Publisher Fees Using Dynamic Revenue Sharing in Order to Route More Transactions Through Its Ad Exchange and Deny Scale to Rival Ad Exchanges Using Header Bidding	86
d)	Google Launches Project Poirot to Manipulate Its Advertisers’ Spend to “Dry Out” and Deny Scale to Rival Ad Exchanges That Use Header Bidding.....	90
e)	Google Imposes So-Called Unified Pricing Rules to Deprive Publishers of Control and Force More Transactions Through Google’s Ad Exchange	101
f)	Google Outright Blocks the Use of Standard Header Bidding on Accelerated Mobile Pages.....	110
g)	Google Replaces Its Last Look Preference from Dynamic Allocation with an Algorithmic Advantage and Degrades Data Available to Publishers	113
V.	Anticompetitive Effects	116
VI.	Relevant Markets	123
A.	Geographic Markets.....	124
B.	Product Markets	124
1.	Publisher Ad Servers.....	124
2.	Ad Exchanges	126
3.	Advertiser Ad Networks	129
VII.	Jurisdiction, Venue, and Commerce	131
VIII.	Violations Alleged	132
IX.	Request for Relief	139
X.	Demand for a Jury Trial.....	140

I. INTRODUCTION

1. An open, vibrant internet is indispensable to American life. But today's internet would not exist without the digital advertising revenue that, as a practical matter, funds its creation and expansion. The internet provides the public with unprecedented access to ideas, artistic expression, news, commerce, and services. Content creators span every conceivable industry; they publish diverse material on countless websites that inform, entertain, and connect society in vital ways. Yet the viability of many of these websites depends on their ability to sell digital advertising space. Just as newspaper, radio, and television organizations historically relied on advertising to fund their operations, today's online publishers likewise rely on advertising revenue to support their activities and reach. But unlike historical media advertising, today's online ads are bought and sold in enormous volumes in mere fractions of a second, using highly sophisticated tools and automated exchanges that more closely resemble a modern stock exchange than an old-fashioned, bilateral contract negotiation for newspaper ad space.

2. Website publishers in the United States sell more than 5 trillion digital display advertisements on the open web each year—or more than 13 billion advertisements *every day*. The sheer volume of these online ads make the offline advertisements of yesteryear pale in comparison. To put these numbers in perspective, the daily volume of digital display advertisements grossly outnumbers (by several multiples) the average number of stocks traded each day on the New York Stock Exchange. The digital display advertising business is also lucrative. Collectively, these advertisements generate more than \$20 billion in revenue per year, just for publishers based in the United States.

3. To meet this demand, sophisticated technological tools, informally known as “ad tech,” have developed to automate advertising matchmaking between two key groups: website

publishers and advertisers.¹ These tools have evolved such that today, every time an internet user opens a webpage with ad space to sell, ad tech tools almost instantly match that website publisher with an advertiser looking to promote its products or services to the website's individual user. This process typically involves the use of an automated advertising exchange that runs a high-speed auction designed to identify the best match between a publisher selling internet ad space and the advertisers looking to buy it.

4. But competition in the ad tech space is broken, for reasons that were neither accidental nor inevitable. One industry behemoth, Google, has corrupted legitimate competition in the ad tech industry by engaging in a systematic campaign to seize control of the wide swath of high-tech tools used by publishers, advertisers, and brokers, to facilitate digital advertising. Having inserted itself into all aspects of the digital advertising marketplace, Google has used anticompetitive, exclusionary, and unlawful means to eliminate or severely diminish any threat to its dominance over digital advertising technologies.

5. Google's plan has been simple but effective: (1) neutralize or eliminate ad tech competitors, actual or potential, through a series of acquisitions; and (2) wield its dominance across digital advertising markets to force more publishers and advertisers to use its products while disrupting their ability to use competing products effectively. Whenever Google's customers and competitors responded with innovation that threatened Google's stranglehold over any one of these ad tech tools, Google's anticompetitive response has been swift and effective. Each time a threat has emerged, Google has used its market power in one or more of these ad

¹ Internet advertisers include businesses, agencies of federal and state governments, charitable organizations, political candidates, public interest groups, and more. The money these advertisers spend on digital advertising creates an important stream of revenue for websites to use in creating, developing, and publishing website content.

tech tools to quash the threat. The result: Google's plan for durable, industry-wide dominance has succeeded.

6. Google, a single company with pervasive conflicts of interest, now controls: (1) the technology used by nearly every major website publisher to offer advertising space for sale; (2) the leading tools used by advertisers to buy that advertising space; and (3) the largest ad exchange that matches publishers with advertisers each time that ad space is sold. Google's pervasive power over the entire ad tech industry has been questioned by its own digital advertising executives, at least one of whom aptly begged the question: "[I]s there a deeper issue with us owning the platform, the exchange, and a huge network? The analogy would be if Goldman or Citibank owned the NYSE."

7. By deploying opaque rules that benefit itself and harm rivals, Google has wielded its power across the ad tech industry to dictate how digital advertising is sold, and the very terms on which its rivals can compete. Google abuses its monopoly power to disadvantage website publishers and advertisers who dare to use competing ad tech products in a search for higher quality, or lower cost, matches. Google uses its dominion over digital advertising technology to funnel more transactions to its own ad tech products where it extracts inflated fees to line its own pockets at the expense of the advertisers and publishers it purportedly serves.

8. Google's anticompetitive behavior has raised barriers to entry to artificially high levels, forced key competitors to abandon the market for ad tech tools, dissuaded potential competitors from joining the market, and left Google's few remaining competitors marginalized and unfairly disadvantaged. Google has thwarted meaningful competition and deterred innovation in the digital advertising industry, taken supra-competitive profits for itself, and

prevented the free market from functioning fairly to support the interests of the advertisers and publishers who make today's powerful internet possible.

9. The harm is clear: website creators earn less, and advertisers pay more, than they would in a market where unfettered competitive pressure could discipline prices and lead to more innovative ad tech tools that would ultimately result in higher quality and lower cost transactions for market participants. And this conduct hurts all of us because, as publishers make less money from advertisements, fewer publishers are able to offer internet content without subscriptions, paywalls, or alternative forms of monetization. One troubling, but revealing, statistic demonstrates the point: on average, Google keeps at least *thirty cents*—and sometimes far more—of each advertising dollar flowing from advertisers to website publishers through Google's ad tech tools. Google's own internal documents concede that Google would earn far less in a competitive market.

10. The United States and Plaintiff States bring this action for violations of the Sherman Act to halt Google's anticompetitive scheme, unwind Google's monopolistic grip on the market, and restore competition to digital advertising.

II. NATURE OF THIS ACTION

11. The seeds for Google's eventual march toward a monopoly in ad tech were sown in the early 2000s, when it capitalized on its well-known search engine to start a profitable search advertising business. In 2000, Google launched **Google Ads** (then called AdWords²), a tool that allowed businesses to buy advertisements that could be seen by Google search users right alongside Google's popular search engine results. Businesses quickly learned the power of

² Over the period addressed by the Complaint, Google has renamed its ad tech products a number of times and has either shifted certain functions between products or combined its products in ways intended to obscure Google's dominance across the ad tech stack.

this instantaneous, highly-targeted advertising technique, and they flocked to Google Ads as a result.

12. By the early 2000s, Google realized that these same advertisers would buy digital advertisements on third-party websites as well. So Google stepped in to profit (as a middleman) on digital advertising transactions having nothing to do with Google or its search engine by creating an advertiser ad tech tool for Google Ads' customers that wanted to buy ad space on third-party websites.

13. But Google was not satisfied with its dominance on the *advertising* side of the industry alone; Google devised a plan to build a moat around the emerging ad tech industry by developing a tool that would be used by website *publishers* as well.

14. Google sought to develop an ad tech tool called a **publisher ad server** that publishers would use to manage their online advertising sales. Google recognized that because publisher ad servers set the rules for how and to whom publisher advertising opportunities are sold, owning a publisher ad server was key to having visibility into, and control over, the publisher side of digital advertising. By controlling the *publisher* ad server on the other end of the transaction, Google could further entrench its *advertiser* customer base by giving advertisers access to more advertising opportunities and pushing more transactions their way.

15. Of course, by becoming the dominant player on both sides of the digital advertising industry, Google could also play both sides against the middle. It could control both the publishers with digital ad space to sell, as well as the advertisers who want to buy that space. With influence over advertising transactions end-to-end, Google realized it could become “the be-all, and end-all location for all ad serving.” The outsized influence it could obtain by having a dominant position on both sides of the industry would give Google the ability to charge supra-

competitive fees and also enjoy an abiding dominance sufficient to exclude rivals from competition. Google would no longer have to compete on the merits; it could simply set the rules of the game to exclude rivals.

16. The only problem with Google's plan was that Google's publisher ad server failed to gain traction in the industry. So, Google pivoted to acquiring the market-leading publisher ad server from an ad tech firm called DoubleClick. In early 2008, Google closed its acquisition of DoubleClick for over \$3 billion. Through the transaction, Google acquired a publisher ad server ("**DoubleClick for Publishers**" or "**DFP**"), which had a 60% market share at the time. It also acquired a nascent **ad exchange** ("**AdX**") through which digital advertising space could be auctioned. The DoubleClick acquisition vaulted Google into a commanding position over the tools publishers use to sell advertising opportunities, complementing Google's existing tool for advertisers, Google Ads, and set the stage for Google's later exclusionary conduct across the ad tech industry.

17. After the DoubleClick acquisition, Google enhanced and entrenched DFP's already-dominant market position. Google internally recognized that publisher ad servers are "sticky" products, meaning that publishers rarely switch because of the high costs and risks involved. As DoubleClick's former CEO observed, "Nothing has such high switching costs. . . . Takes an act of God to do it." Thus, in order to lock more publishers into DFP and to reinforce its stickiness, Google forged an exclusive link between Google Ads and DFP through the AdX ad exchange. If publishers wanted access to exclusive Google Ads' advertising demand, they had to use Google's publisher ad server (DFP) and ad exchange (AdX), rather than equivalent tools offered by Google's rivals. In effect, Google positioned itself to function simultaneously as buyer, seller, and auctioneer of digital display advertising.

18. Google's strategy paid off. This arrangement has had a profound effect on the evolution of digital advertising. First, it tilted the industry in Google's favor, driving publishers to adopt and stay on Google's DFP publisher ad server in order to have access to Google Ads' advertiser demand. Second, it cut off the possibility that Google Ads' advertiser spending could sustain, or encourage the entry of, a rival ad exchange or publisher ad server by providing critical advertising demand. For the vast majority of webpage publishers, this arrangement made DFP the only realistic publisher ad server option. Indeed, by 2015, Google estimated that DFP's publisher ad server market share had grown to a remarkable 90%. Google's durable monopoly over the publisher ad server market has allowed it to avoid innovation and competition by controlling the very rules by which the game is played. As a result, other publisher ad servers have left the market altogether, refocused on related markets, or faded into insignificance; no new publisher ad servers have entered the market.

19. Around the same time that Google tied its exclusive Google Ads' advertiser demand to its publisher ad server (DFP) through AdX, Google took two additional steps to make it more difficult for rivals to compete.

20. *First*, Google configured Google Ads to bid on Google's AdX ad exchange in a way that actually *increased* the price of advertising, to the benefit of publishers and the detriment of Google's own advertiser customers. As one Google employee observed, Google Ads was effectively sending a "\$3bn yearly check [to publishers] by overcharging our advertisers to ensure we're strong on the pub[lisher] side." In the short-term, this conduct locked publishers into Google's publisher ad server by providing them a steady stream of intentionally-inflated prices for certain inventory, at the cost of Google's own advertiser customers. But in the long run, Google's actions harmed publishers as well by driving out rival publisher ad servers and

limiting competition in the publisher ad server market. In effect, Google was robbing from Peter (the advertisers) to pay Paul (the publishers), all the while collecting a hefty transaction fee for its own privileged position in the middle. This conduct turned the entire purpose of the digital advertising industry on its head. Rather than helping to fund website publishing, Google was siphoning off advertising dollars for itself through the imposition of supra-competitive fees on its platforms. A rival publisher ad server could not compete with Google's inflated ad prices, especially without access to Google's captive advertiser demand from Google Ads.

21. *Second*, Google used its captive advertiser demand to thwart legitimate competition by giving its AdX ad exchange an advantage over other ad exchanges through a mechanism known as dynamic allocation. Dynamic allocation was a means by which Google manipulated its publisher ad server to give the Google-owned AdX (and only AdX) the opportunity to buy publisher inventory before it was offered to any other ad exchange, and often to do so at artificially low prices. Google also programmed DFP, its publisher ad server, to prevent publishers from offering preferential terms to other ad exchanges or allowing those exchanges to operate in the same way with DFP. Google knew that dynamic allocation would inevitably steer advertising transactions away from rivals, denying them critical scale needed to compete, and would advantage AdX, where Google could extract the largest fees. Google's scheme predictably reinforced publishers' dependence on both AdX and DFP. Publishers were effectively precluded from using rival ad servers or ad exchanges that might better suit their needs while Google was given a free pass from having to compete on the merits with those rivals.

22. By at least 2010, other ad tech companies had recognized that Google's platforms were not working in the best interest of publishers, and they attempted to develop innovative

technologies to introduce more competition. Some companies began offering “yield management” functionality that helped publishers identify on a real-time basis better prices for their inventory outside of Google’s products. Google recognized that yield managers posed a major threat to the increasingly closed system Google sought to establish, in which only its ad exchange was able to compete based on real-time pricing. So, in response, Google employed a familiar tactic: acquire, then extinguish, any competitive threat.

23. In 2011, Google acquired AdMeld, the leading yield manager, folded its functionality into Google’s existing products, and then shut down its operations with non-Google ad exchanges and advertiser tools. Google soon thereafter changed its AdX contract terms to prohibit publishers from using any other platform (such as another yield manager) that would force AdX to compete in real time with other ad exchanges. As a Google product manager wrote: “Our goal should be all or nothing – use AdX as your [exchange] or don’t get access to our [advertising] demand.” Unsurprisingly, this unabashed, anticompetitive conduct had a profound effect on the market, denying rival ad tech competitors the scale necessary to compete and depriving publishers the benefits of free market competition and real choice.

24. Not long after, in 2013, Google launched **Project Bernanke**, a secret scheme to manipulate the bids that Google Ads submitted into Google’s ad exchange, AdX, in order to win more competitive transactions and solidify AdX’s dominance in the industry. Project Bernanke allowed Google to suppress competition by preventing rival ad exchanges from achieving the transaction volume and scale necessary to compete. Unless another ad exchange developed both its own unique source of captive advertiser demand—where it could potentially manipulate advertiser bids—and a widely-adopted publisher ad server—where it could see the same advertising inventory and bid data as Google—competition on the same terms as Google was

nearly impossible. Once again, by controlling all sides of the ad tech industry, Google has been able to manipulate the system in ways unique to itself so that, in the end, it did not have to compete on the merits for customers and volume.

25. Publishers and competing ad tech providers, increasingly wary of Google's bullying behavior, have continued to look for new ways to circumvent Google's dominance. Between 2012 and 2013, market participants began using a technique called "**header bidding**" as a partial workaround to Google's self-preferential algorithms and ad tech restrictions. As one Google employee explained, "Publishers felt locked-in by dynamic allocation in [Google's ad server] which only gave [Google's ad exchange] the ability to compete, so HB [header bidding] was born."

26. Publishers used header bidding to take back some degree of power over their own advertising transactions. They inserted header bidding computer code onto their own websites to allow non-Google advertising exchanges an opportunity to bid for advertising inventory before Google's hard-coded preferences for its own ad exchange were triggered. Header bidding allowed publishers to ensure that multiple advertising exchanges—not just Google's AdX—could bid on their inventory, thereby increasing the chances that they could find the best match.

27. Google has refused to tolerate this new form of competition, even though it has acknowledged in internal emails that header bidding had grown naturally out of Google's being "unwilling[] to open our systems to the types of transactions, policies and innovations that buyers and sellers wish to transact." Indeed, Google privately admitted that "header bidding and header wrappers are BETTER than [Google's platforms] for buyers and sellers," and that increased competition between AdX and publishers using header bidding would increase publisher revenues by 30 to 40%, and would provide additional transparency to advertisers. Not

only would header bidding enable rival exchanges to compete more effectively against Google's ad exchange, it might also allow them or others to enter the publisher ad server market if Google no longer had exclusive access to unique advertiser demand.

28. Google executives described header bidding as an "existential threat." They worried that wider adoption of header bidding practices could lead to Google's ad exchange having to compete with other ad exchanges on a level playing field, where Google could no longer set the rules in its own favor. If that were to happen, those rival ad exchanges might actually succeed in eroding, or even breaking up, Google's advertiser juggernaut, and the entire industry could be opened up for competition. Google feared the worst: the entire moat of anticompetitive protections that Google had built around the ad tech industry could be breached.

29. Faced with this "existential" threat, Google sought to stem the rising tide toward header bidding by promoting a Google-friendly analog of header bidding that Google deceptively titled "Open Bidding." Google has promoted Open Bidding as an answer to the industry's call for wider participation by rival ad exchanges and increased competition. In fact, Open Bidding was a Trojan Horse that Google used to further cement its own monopoly power.

30. As a condition to using Google's Open Bidding, Google has required that publishers and participating ad exchanges give Google visibility into each auction (including how rival exchanges bid), allow Google to extract a sizeable fee on every transaction (even where another exchange won), and limit the pool of advertisers allowed to bid in the auctions. In doing so, Google's ad exchange has retained a guaranteed seat in every auction, regardless of whether Google's ad exchange offers the best match between advertisers and publishers.

31. Google also sought to co-opt what it perceived to be its two biggest threats (Facebook and Amazon) into Open Bidding. In internal documents, Google concluded that while

it “[c]annot avoid competing with FAN [Facebook],” it could, through a deal with Facebook, “build a moat around our demand.” Internal documents recommending a deal with Facebook revealed Google’s primary motive: “[f]or web inventory, we will move [Facebook’s] demand off of header bidding set up and further weaken the header bidding narrative in the marketplace.” Thus, for these reasons, Google ultimately agreed to provide preferential Open Bidding auction terms to Facebook in exchange for spend and pricing commitments designed to push more of Facebook’s captive advertiser spend onto Google’s platforms. Google sought to head off Amazon’s investment in header bidding technology with a similar offer, albeit without the same success.

32. Google also adjusted its auction mechanisms across its ad tech products to divert more transactions to itself and away from rivals that might deploy header bidding. On the publisher side, Google allowed AdX—and only AdX—to change its auction bid by altering Google’s own fee after seeing the price to beat from another exchange.

33. On the advertiser side, Google first considered outright blocking its advertiser buying tool from buying inventory made available via header bidding. The goal: “dry out HB [header bidding].” When Google decided that strategy would be too costly for Google, it pivoted to a different and more insidious strategy with the same effect.

34. Google recognized that “instead of stop[ping] bidding on HB [header bidding] queries, we could bid lower on HB queries,” and win the same impressions on Google’s ad exchange instead. No rival exchange was in a position to compete with this strategy because no rival had the scale necessary to compete against the industry giant, especially considering the built-in advantages that Google afforded its own ad exchange and publisher ad server. Google, and Google alone, had control over both the leading source of advertiser demand and the

dominant publisher ad server. So, Google programmed its advertiser buying tool to advantage its ad exchange.

35. Google's bidding strategy on header bidding transactions proved remarkably effective in stunting the growth of header bidding, but Google still worried that its moat was not fully secure. Google learned that some publishers were using price controls within Google's own DFP publisher ad server to sell advertising inventory to rival exchanges outside of Google's closed-wall system, even in instances where Google's own AdX exchange had offered to pay more for the inventory. Publishers did so for a variety of reasons, including considerations related to ad quality, volume discounts, diversification of demand sources, data asymmetries, or other factors.

36. When Google identified this threat, it simply removed the feature from DFP and instead imposed competition-stifling Unified Pricing Rules. Under these new rules, publishers could no longer use price floors to choose rival exchanges or other buyers over AdX or Google Ads, no matter the reason. Google effectively took away their own customers' right to choose what buyer or ad exchange best suited their needs. In doing so, Google once again bought itself a free pass on competition.

37. Google's exclusionary, anticompetitive acts have severely weakened, if not destroyed, competition in the ad tech industry. In decision after decision, year after year, Google has repeatedly done what was necessary to vanquish competitive threats, including by enacting policies that took choices away from its own customers. And despite what Google may claim, it did not do so to protect the privacy interests of Google users. Indeed, Google intentionally

exploited its massive trove of user data to further entrench its monopoly across the digital advertising industry.³

38. Due to Google's conduct, ad tech tools that should have evolved to better serve website publishers and advertisers in a competitive environment have instead evolved to serve the interests of Google alone, to the detriment of Google's own customers. The results have been catastrophic for competition. Today, major website publishers have a single viable choice for publisher ad servers—Google's DoubleClick for Publishers. Google routes transactions from its publisher ad server to its more expensive ad exchange—AdX—and away from rival platforms, all of which are less than a quarter of AdX's size.

39. Advertisers and publishers, the key players in this market, have had scant visibility into the scope and extent of Google's anticompetitive conduct. As the lone conflicted representative of both buyers and sellers, Google has created a deliberately-deceptive black box where Google sets the auction rules to its own advantage. Diminished competitive pressure has reduced Google's incentive to innovate, and Google's control of these key ad tech tools has inhibited rivals' ability to introduce efficiency-enhancing innovations. Publishers and advertisers suffer from reduced competition for both ad tech products and advertising inventory. Google's conduct undermines the very purpose of digital advertising in the first place: to achieve optimum terms and pricing for digital advertisements so website publishers can continue to serve their

³ At the time of the DoubleClick acquisition, Google's privacy policies prohibited the company from combining user data obtained from its own properties, e.g., Search, Gmail, and YouTube, with data obtained from non-Google websites. But in 2016, as part of Project Narnia, Google changed that policy, combining all user data into a single user identification that proved invaluable to Google's efforts to build and maintain its monopoly across the ad tech industry. Over time, Google used this unique trove of data to supercharge the ability of Google's buying tools to target advertising to particular users in ways no one else in the industry could absent the acquisition of monopoly—or at least dominant—positions in adjacent markets such as Search.

vital purposes in society. Indeed, Google's own documents show that Google has siphoned off *thirty-five cents* of each advertising dollar that flows through Google's ad tech tools:



Fig. 1

40. The cumulative impact of Google's anticompetitive conduct is more than simply the sum of each harm Google has caused. As new threats have arisen, Google has spread its actions across wide-ranging ad tech products knowing the synergistic, multiplier effect that its actions would have across the industry. Because Google has such a powerful hand in each aspect of the ad tech industry, it alone has the power to use and deploy hidden levers to manipulate the overall system to its advantage.

41. It is critical to restore competition in these markets by enjoining Google's anticompetitive practices, unwinding Google's anticompetitive acquisitions, and imposing a remedy sufficient both to deny Google the fruits of its illegal conduct and to prevent further harm to competition in the future. Absent a court order for the necessary and appropriate relief, Google will continue to fortify its monopoly position, execute its anticompetitive strategies, and thwart

the competitive process, thereby raising costs, reducing choice, and stifling innovation in this important industry.

III. DISPLAY ADVERTISING TRANSACTIONS

42. When an internet user opens a website, a complex series of transactions—nearly instantaneous and invisible to the user—determines which ad to show to that user in each available ad space on the webpage. The set of technological tools that connect website publishers selling advertising opportunities to the advertisers wishing to buy those advertising opportunities (“ad inventory”) is referred to as **ad tech**. Below is a schematic depicting some of the important ad tech tools used in online digital advertising:

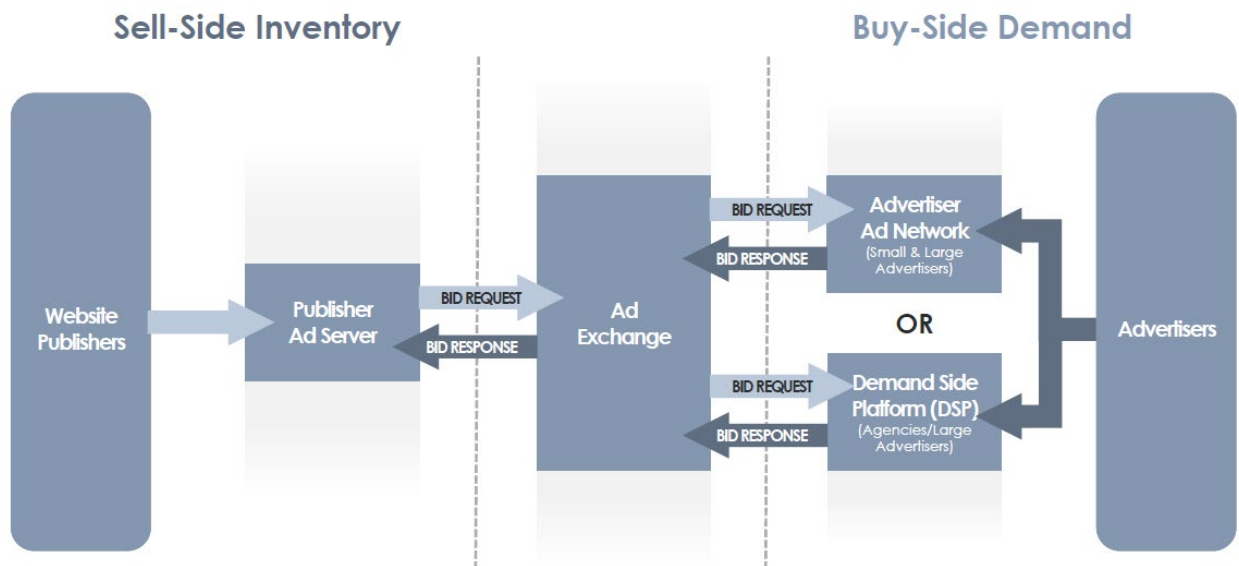


Fig. 2

A. *How Ad Tech Tools Work*⁴

43. The content creator or owner of a website is called a **publisher**. Each website can be programmed by its publisher to create slots where ads can be displayed. A graphical ad

⁴ The process described herein governs the sale of display ads on the “**open web**,” meaning websites whose inventory is sold through ad tech intermediaries that offer inventory from

displayed on a website that is viewed in an internet browser is called a **display ad**. A display ad may contain images, text, or multimedia. A single display ad shown to a single user on a single occasion is called an **impression**.

44. An ad tech transaction begins when a user opens a website. While the website's content loads, the website uses a **publisher ad server** to select which ads will fill each ad slot on the page. The publisher ad server is an ad tech tool that evaluates potential ads from different advertising sources and applies a decision-making logic to determine which ad will be displayed to the user opening the website. Since 2008, Google has owned the industry's leading publisher ad server, **Google Ad Manager**, which is often still referred to by its former name, **DoubleClick for Publishers** ("DFP").

45. For a typical medium-to-large website, the publisher ad server first determines whether the ad spaces on the webpage opened by the user have already been sold to a specific advertiser directly by the publisher. Such **direct sales** result from one-on-one negotiations between website publishers and advertisers and typically involve premium ad placements (e.g., ads at the top of a webpage) that command the highest prices from advertisers. For any ad space not filled through direct sales, the publisher ad server then tries to sell the ad space through indirect sales channels. **Indirect sales** allow publishers to sell remaining or "remnant" ad space

multiple websites. Some websites, especially social media companies like Facebook and Snapchat, operate under a different "closed web" (or "walled garden") model in which inventory is sold directly to individual advertisers using a proprietary tool employed by that website. Other types of advertising distinct from open web display advertising include search ads (e.g., sponsored results in a search engine), video ads (e.g., commercials that play before, during, or after a streaming video), and mobile app ads (e.g., ads shown within a game or other non-browser app downloaded from an app store to a user's mobile device). The focus of this Complaint is on Google's anticompetitive conduct in the market for open web display advertising transactions.

(i.e., space not sold through direct sales). Many website publishers, especially smaller ones, only sell ad space through such indirect sales.⁵

46. Indirect sales are typically made via a series of interactions between ad tech tools. These technologies allow website publishers and advertisers to transact through lightning-fast automated processes, known as **programmatic** buying. Today, most programmatic transactions take place on an ad exchange. An **ad exchange** (sometimes called a supply-side platform or SSP) is a software platform that receives requests—often from a publisher ad server—to auction ad impressions on a particular webpage. The ad exchange solicits bids on the impression from advertiser buying tools, chooses the winning bid, and transmits information on the winning bid back to the publisher ad server. Google presently owns the industry’s leading ad exchange, called **AdX** (now packaged as part of **Google Ad Manager**).

47. When a publisher ad server sends an auction request to an ad exchange, the publisher ad server provides certain information about the impression for sale. This can include information about the website itself, the ad space on the webpage (e.g., where the ad is placed), and the user that will view the impression.⁶ After receiving this information from the publisher ad server, the ad exchange may supplement the information with any additional information the ad exchange might independently have about the user viewing the ad, including information about the user’s browsing history, location, and age.⁷ The ad exchange then transmits the bid

⁵ For both direct and indirect sales, ad impressions are generally priced on a **CPM** basis, referring to cost-per-thousand (in Latin, “mille”) per impression. For example, an impression with a \$1 CPM would cost \$0.001, or one-tenth of a cent.

⁶ Because the publisher ad server historically transmitted this information to the ad exchange, the publisher ad server controlled what information was sent to prospective advertisers and in what form.

⁷ Information concerning the user’s location and browsing history can be gleaned through “cookies” set in place by the user’s web browser. These cookies allow the web browser to collect

request, along with information gathered about the user and the website, to various advertiser buying tools, described below. The detailed information concerning the user's location and browsing history is highly valuable to advertisers because it helps advertisers assess the value of the particular impression to its overall advertising campaign. For example, if the information tells a particular retail advertiser that the user had previously browsed that retailer's website but did not complete a sale, then that retailer may be willing to pay a premium for the particular impression.

48. Advertisers receive and respond to bid requests using **advertiser buying tools**. These advertiser buying tools assist advertisers with connecting to ad exchanges, selecting impressions to bid on, submitting bids, and tracking the purchased impressions against the advertiser's advertising campaign goals.

49. Large ad buyers, such as major ad agencies or large businesses, frequently use a type of advertiser buying tool called a **demand side platform**. Demand side platforms provide sophisticated and customizable tools that allow the ad agency or business to manage their advertising purchases. Advertisers using demand side platforms have extensive control over where and how they bid for ad inventory. They often use their own data, or data purchased from other entities, to target particular users for their ad campaign. Google owns the United States' leading demand side platform, **Display & Video 360 ("DV360")**.

50. Smaller advertisers often rely on a type of advertiser buying tool with fewer, simpler options that are less customized. These advertiser buying tools are called **advertiser ad**

information about a user's internet location and browsing history which can then be passed along, or sold, to interested parties.

networks.⁸ Today, most ad networks bid for and buy advertising space on an impression-by-impression basis, submitting bids alongside other ad networks and demand side platforms.

Advertiser ad networks offer a self-service, easy-to-use technology solution, which as a practical matter is the only viable option for smaller advertisers, advertisers that prefer a simple “hands-off” approach, or advertisers that need the ad network’s targeting data to buy ads effectively.

Google offers the industry’s leading ad network, **Google Ads**.

51. Most ad networks, including Google Ads, are a “black box” to advertisers. Advertisers have almost no control over the process by which the ad network bids for impressions. Nor do the networks provide advertisers with information about how or why the network bids for particular impressions on particular websites at particular times. Most ad networks charge advertisers primarily on a “cost per click,” or “CPC” basis. The advertiser thus has no insight into how much the ad network spent to purchase a particular impression; the advertiser is charged a fee only when an internet user clicks on the ad. Google’s ad network, Google Ads, sets this fee based on the actual cost incurred to buy advertising inventory plus a markup. This prevents Google’s advertising customers from knowing how much Google is charging them, over and above Google’s costs, for the inventory.

52. These ad networks are particularly important to businesses that do not have the expertise, advertising budget, or targeting data required for a demand side platform to be a viable option. Ad networks are also critical to website publishers. These ad networks are the only way for publishers to reach and sell ad space to smaller businesses that rely exclusively or primarily

⁸ These advertiser networks are referred to as “networks” because they originally operated on a network model whereby the ad network would agree to buy a portion of a publisher’s advertising space in bulk at a pre-set price. The ad network would then distribute the publisher’s advertising space among a network of advertisers. The prices charged to those advertisers were not necessarily derived from the bulk price the network paid to acquire the space.

on ad networks to buy ad space. Further, the type of advertising space these ad networks seek to purchase from website publishers is often distinct from the advertising space sought by other advertising tools. That is because the advertisers using these networks often have unique advertising objectives. Further, these ad networks, and in particular Google Ads, have access to unique user data that allow them to target very specific advertising opportunities.

53. The flow of display ad transactions through these platforms—collectively called the **ad tech stack**—is depicted again below.

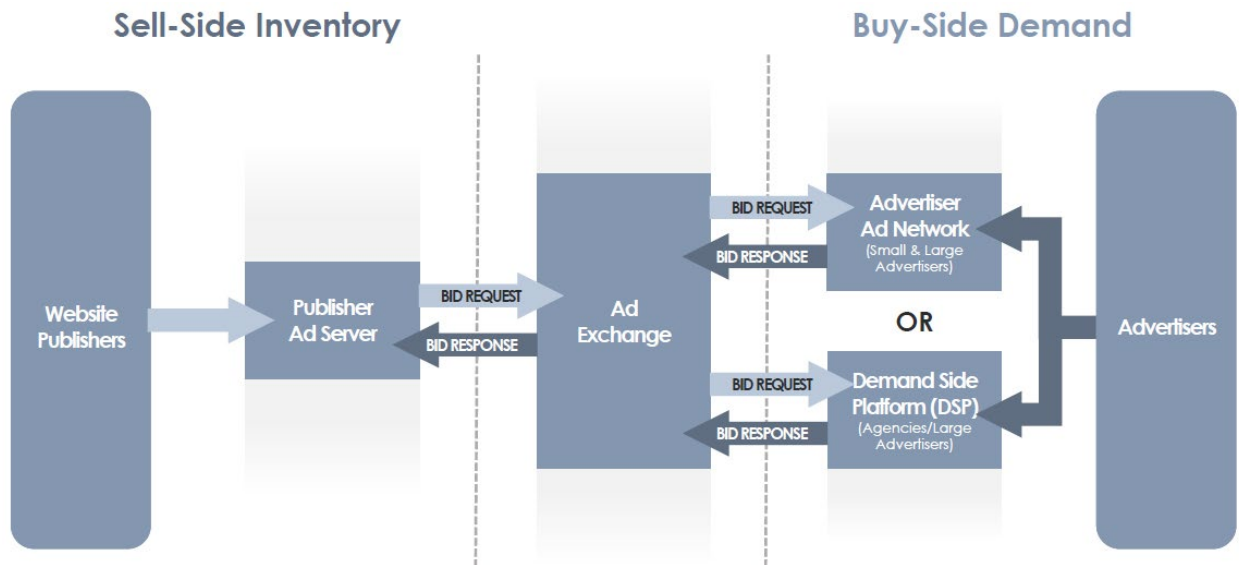


Fig. 3

54. The publisher ad server is referred to as the “**sell-side**.” The advertiser buying tools are referred to as the “**buy-side**.” Impressions offered for sale by publishers are referred to as publisher “**inventory**” and advertisers’ interest in buying impressions is referred to as advertiser “**demand**.”

55. Whether the advertiser uses a demand side platform or an ad network as its advertiser buying tool, the tool evaluates the bid request received from the ad exchange and, if the impression meets the advertiser’s criteria (e.g., targeted audience, website category), the tool

determines an amount to bid on the impression. Because each impression is filled within fractions of a second while the website loads for the user, an advertiser could never evaluate each impression individually. Instead, advertisers rely on these automated advertiser buying tools to evaluate impressions and bid on their behalf based on parameters pre-configured by the advertiser ahead of time. The advertiser buying tool then sends its highest bid for the impression—as calculated by the tool—back to the ad exchange for consideration.

56. After receiving bids from multiple advertiser buying tools, the ad exchange holds an auction to determine the winning bidder. Historically, most ad exchanges ran a second-price auction in which the winning bidder paid a price one cent higher than the bid of the second-highest bidder. Today, however, most ad exchanges run first-price auctions where the highest bidder simply pays the price of its winning bid. The ad exchange sends information about its winning bid back to the publisher ad server, which evaluates the ad exchange's bid under a set of rules defined by the publisher ad server. The publisher ad server then makes the final decision regarding which ad to “serve” to the user. The publisher ad server sends a message to the winning advertiser to provide the content of the ad to be displayed.

B. How Ad Tech Intermediaries Get Paid

57. Once the winning bid has been chosen, the advertiser pays the website publisher for the impression, but a portion of the payment is retained by each intermediary along the way as payment for its services. The advertiser buying tool and the ad exchange supplying the winning bid each collect a portion of the purchase price for the impression, which is referred to as a “**revenue share**” or “**take rate**.” The publisher ad server generally charges the publisher a fee based on the number of impressions served. Unlike a revenue share, the publisher ad server fee typically does not vary based upon the price paid for each particular impression.

58. The total percentage of advertiser spend extracted by ad tech intermediaries can have a substantial impact on the revenue website publishers earn from advertising and on the return on investment that advertisers receive from their advertising campaigns. But this percentage is typically not fully transparent to advertisers or publishers; some fees are disclosed only to publishers or advertisers while other fees are obscured or not disclosed at all. According to Google's internal documents, when a transaction passes through each of Google's ad tech tools (including Google's campaign manager product, which helps advertisers manage ad content and track campaign spending), Google estimates that it gets to keep about 35% of every dollar spent on digital advertising (as shown in Figure 4 below).

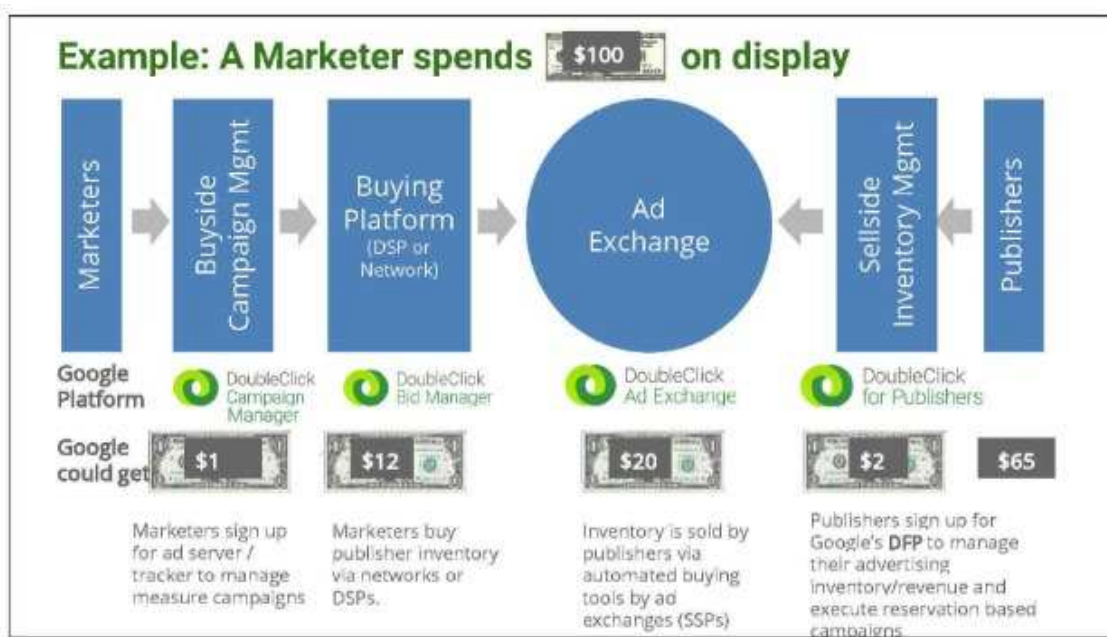


Fig. 4

59. These technology platforms have provided essentially the same services for over a decade. During that time, Google's monopoly positions and the restrictions it has imposed across these technologies have diminished the incentive and ability for Google or others to innovate. This reduced innovation is compounded by high prices: despite publishers' and advertisers'

interests in reducing the amount of advertising spending siphoned off by intermediaries, Google's take rate has remained remarkably stable over time. In particular, Google has consistently charged a roughly 20% fee for impressions bought through its ad exchange, the link in the chain where the highest fees are charged.

C. How Publishers and Advertisers Select Ad Tech Tools

60. Publishers and advertisers try to optimize their use of ad tech to meet their revenue or advertising goals. As a general matter, publishers use only one publisher ad server to manage ad inventory in order to avoid discrepancies in tracking revenue or impressions and to minimize the burden of having employees oversee two largely duplicative systems. Ultimately, there can only be one publisher ad server acting as the final decision-maker as to which advertisement will fill each impression.

61. Sizeable publishers generally prefer to offer their inventory for sale through more than one ad exchange (a practice called “**multi-homing**”). This increases the likelihood that an advertiser on one or more of the ad exchanges will be able to “match” the advertising opportunity offered by the publisher to a user or category of user that an advertiser particularly values and therefore is willing to compete to buy. When publishers are able to offer their inventory for sale through multiple ad exchanges simultaneously, it causes ad exchanges to compete with each other to provide the best “match” or the lowest revenue share. However, there are integration, contracting, and other costs associated with the publisher adding each additional ad exchange.

62. Likewise, advertisers often connect with multiple ad exchanges through their advertiser buying tools, hoping that exposure to as much advertising inventory as possible will increase the likelihood of reaching the advertisers' intended targets for their advertising campaigns at the lowest cost. Using multiple ad exchanges also allows advertisers to compare

performance between ad exchanges. Similarly, when advertisers are able to freely multi-home among ad exchanges, it forces ad exchanges to compete with each other to provide advertisers the best return on their advertising expenditures.

63. Although there are a number of factors that advertisers consider when deciding which ad exchanges and/or ad buying tools to use, one key driver is access to especially valuable advertising inventory. Some ad tech products can be used to buy or sell both open web display advertising—the focus of this Complaint—as well as other types of advertising, such as advertising inventory that is “owned-and-operated” (“O&O”) by the company offering the ad tech product.

64. For example, some of Google’s ad tech products allow advertisers to buy both open web display advertising on third-party websites as well as advertising on Google’s O&O properties. Google’s O&O properties include several market-leading sources of non-open web display advertising inventory, such as Google Search, YouTube, Gmail, and Android’s Google Play Store, among others. Advertisers and advertising agencies looking to advertise on these O&O properties often must adopt at least one of Google’s advertising tools to do so effectively. For example, many larger advertisers and ad agencies seeking to promote their brands through online video advertising on the market-leading YouTube website generally must use Google’s advertising tools to do so; so for them, as well, adoption of Google’s ad tech tools is considered a must.

65. If an advertiser or advertising agency believes it needs Google’s tools for purposes of Google O&O advertising, it is less likely to adopt another buying tool—or tools—to advertise on the open web. Among other considerations, the adoption of multiple ad tech tools typically costs more (in time and money) and limits the ability of the ad tech tools to share

important performance data across these tools. As a result, companies with especially valuable O&O advertising—such as Google—may be able to take advantage of existing, sizeable advertising bases already locked into their advertising tools.

D. Why Scale and the Resulting Network Effects are Necessary to Compete in Ad Tech

66. Scale is a critical factor in the long-term success of each of the key products that comprise the ad tech stack. Scale and related network effects are cumulative; they reinforce market power for incumbents and raise barriers to entry and competition for nascent and smaller rivals. There are at least three important dimensions of scale at play in online digital display advertising.

67. *First*, scale in ad tech means having a significant number and variety of publishers or advertisers using a particular ad tech product. For example, an ad exchange that has significant scale enjoys large numbers and varied types of (i) publisher advertising inventory, on the one hand; and (ii) advertisers that bid through the ad exchange, on the other hand. This scale is key to attracting both publishers and advertisers to the ad exchange because ad exchanges are characterized by strong network effects (meaning that the value of an ad exchange to its users increases as more users adopt the tool). An ad exchange with access to more inventory—especially more sought-after inventory—will be more attractive to advertisers. Likewise, an ad exchange with more advertisers—and more unique advertisers—will be more attractive to publishers. This aspect of scale plays out in similar but less pronounced ways for publisher ad servers. For example, larger and more valuable inventory justify an ad exchange incurring the cost to integrate with a particular ad server. Publisher ad servers are also relatively more expensive to build and relatively less expensive to run, so a larger publisher base allows the publisher ad server to spread the fixed costs over more publishers. With respect to advertising

buying tools, more advertisers and more overall advertising spend will attract publishers to a particular tool. Moreover, to the extent that an advertiser buying tool has access to data from a related sell-side product, the advertiser buying tool can gain unique targeting abilities.

68. *Second*, scale includes the number and quality of impressions that publishers have offered for bidding through the ad tech product, the number of bids advertisers have made, and the number of transactions that have been completed—as well as the associated revenue for those transactions. The more business the ad tech provider has done, the more data that provider has, and the greater the ability the provider has to increase the value of its services. For example, an ad tech provider that is able to see a larger swath of advertising inventory made available for auction will have greater insights into the universe of inventory available, and can adjust—or suggest adjustments to—its customer’s bidding behavior accordingly. Additionally, an ad tech provider that is able to see at scale who ultimately buys or bids on inventory and at what prices can create bidding strategies that can be used to predict more accurately future auctions for similar inventory. For example, the ability to observe the depth and distribution of bids for different advertising inventory can provide valuable data on how demand might change based on price and other factors. In addition, data concerning advertisers’ buying strategies, and how all of this information changes over time, is incredibly useful. Without access to this type of inventory, bidding, and transaction information at scale, an ad tech provider is less able to offer a competitive ad tech tool to publishers or advertisers.

69. *Third*, scale includes the depth of targeting data that an ad tech product has available and can use to identify the most valuable matches between particular pieces of publisher inventory and advertisers. This aspect of scale in the ad tech ecosystem is influenced both by an ad tech provider’s access to relevant targeting data from seeing and winning more

digital advertising transactions (which can provide important information on an internet user's characteristics and behavior) as well as from other parts of its business (e.g., Google's access to website contextual data and detailed user profiles on its customers using Search, Chrome, Android, or Gmail).

70. The ability of an ad tech product to achieve scale along these dimensions is important to its long-term success. For an ad exchange, increasing publisher inventory and advertiser demand, understanding the likely bid landscape based on prior consummated transactions, and having access to detailed user targeting and contextual data all increase the ad exchange's chances of being the supplier of the advertiser bid ultimately selected by the publisher ad server. This is key because ad exchanges only collect a revenue share on *winning* bids—even though the ad exchange incurs costs (for personnel, equipment, and processing power) for every bid request and response, whether won or lost. An ad exchange lacking sufficient access to these various dimensions of scale may not be able to compete effectively, innovate, or even operate.

E. How Multi-Homing Enables Competition in the Ad Tech Stack

71. The purpose of the ad tech stack is to bring together publishers and advertisers. Publishers benefit when there are more advertisers to bid on their inventory, and advertisers benefit when there are more impressions available to buy. As a result, the various markets that make up the ad tech stack exhibit strong “indirect network effects,” i.e., the value of the services provided by these ad tech tools increases as the number of participants on both sides of the product increases.

72. Additionally, because each possible advertising opportunity (or impression) is unique based on a variety of factors (e.g., the identity of the user, the substance of the website, the location on the webpage), the value of a particular impression opportunity can vary

significantly across advertisers. For example, a banner ad at the top of an automotive website would be highly valuable to a car dealership located in the same zip code as the user; that same banner ad space would be less valuable to a home improvement store located in another state. Digital advertising technology, when operating in a healthy, competitive environment, attempts to create the most value for its customers by matching publisher advertising opportunities with the advertisers willing to pay the most for them. By multi-homing across ad exchanges, both website publishers and advertisers are able not only to seek the best possible match for a given advertising opportunity, they are also able to contribute to, and benefit from, competition more generally.

73. Ad exchanges compete for publisher inventory and advertiser demand at two distinct but related levels. First, they compete for adoption by publishers and advertisers, i.e., the opportunity to see a publisher's inventory or submit an advertiser's bid. Second, once an exchange has been adopted, it competes with other exchanges to win the ability to process a particular advertising transaction (i.e., to win individual advertising auctions). At both levels of competition, ad exchanges compete not only on price but also on quality and access. Generally, an ad exchange with more advertisers will be more valuable to publishers, and vice versa. When both sides in a market single-home (i.e., only connect with a single ad exchange), sellers (publishers) tend to flock to the ad exchange with the most buyers (advertisers), all else being equal. Advertisers likewise prefer the ad exchange that has the most advertising inventory from publishers. Google's dominance of scale on both sides of the ad tech stack thereby strengthens Google's dominance overall in the industry and weakens its rivals' ability to compete. Conversely, when participants on both sides actively multi-home, there may be multiple exchanges that offer access to the other side of the market, applying competitive pressure to

decrease fees or increase quality in order to win business. Thus, actions that impair the ability of one or both sides to multi-home are invariably corrosive to competition.

IV. GOOGLE’S SCHEME TO DOMINATE THE AD TECH STACK

74. Over the past fifteen years, Google has acquired and maintained mutually reinforcing monopoly positions in tools across the ad tech stack. Google’s scheme has involved a range of conduct, whereby it—often surreptitiously—has wielded its market power in various ad tech tools to undermine attempts by publishers, advertisers, and rivals to introduce more competition for digital advertising transactions. Individually and in the aggregate, Google’s anticompetitive acts have deprived rivals of critical scale and contributed to Google’s dominance by erecting substantial barriers to entry and competition.

75. Google also has used its dominant position time and again to prevent publishers—its own customers—from efficiently and effectively multi-homing across ad exchanges, and to prevent rival ad tech providers from deploying technology that would have improved the process by which advertisers and publishers find the best advertising matches in real time for each impression. In the face of potential competitive threats, Google has resisted innovation and chosen not to compete on the merits. Instead, it has used acquisitions and market power across adjacent ad tech markets to quash the rise of rivals, tighten its control over the manner and means through which digital advertising transactions occur, and prevent publishers and advertisers from

working effectively with Google's rivals. As the figure below demonstrates, Google's dominance across the ad tech industry is unparalleled.

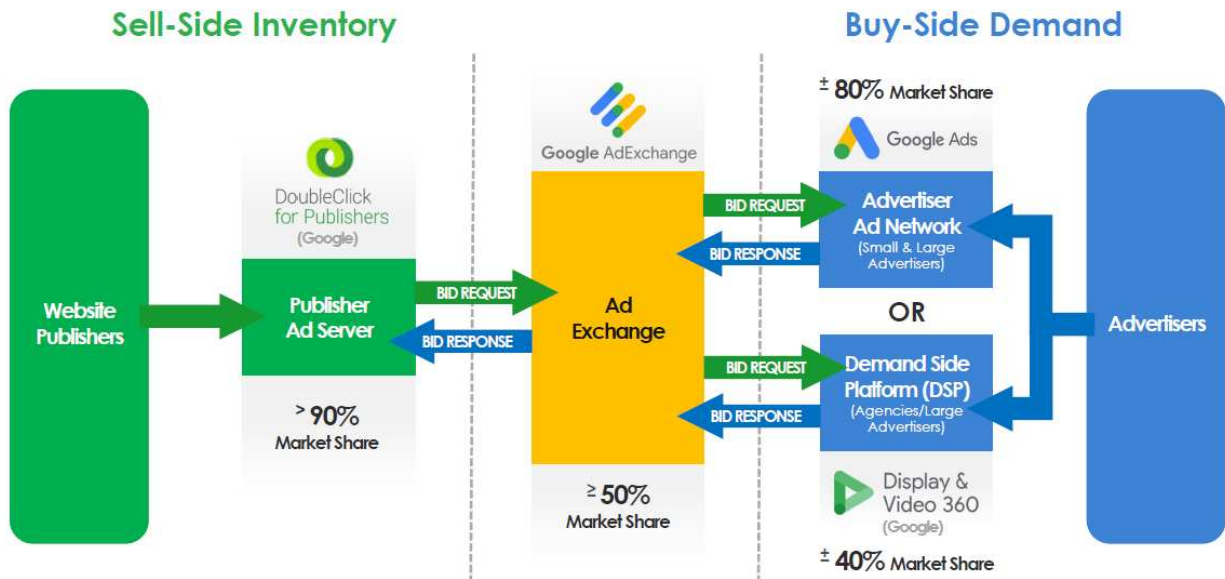


Fig. 5

A. Google Buys Control of the Key Tools that Link Publishers and Advertisers

76. Google entered display advertising on the back of its early strength in search and search advertising. In 2000, Google launched **Google Ads** (then called “AdWords”), a self-service buying tool for advertisers. At the time, advertisers could use Google Ads to purchase advertising on the webpage displaying Google search results.

77. As Google's search engine dominance grew, it attracted large numbers of small and large businesses that considered advertising on Google's search results page to be critical to reaching customers searching for their products or services. After amassing this pool of advertisers, Google realized it could not only sell them advertising space on Google's search results page, but also step in as an intermediary to sell them advertising space on non-Google websites as well. Thus in 2003, Google changed the default setting on Google Ads so that businesses were automatically opted into using Google Ads to advertise on third-party websites

through what became known as **Google Display Network**, or “**GDN**.” Today, Google Ads has grown to represent over two million advertisers, spending about \$11 billion worldwide on open web display inventory per year. Google Ads is a substantial, unique source of advertising demand and revenue for publishers.

78. In 2006, Google found itself without sufficient access to non-Google premium advertising inventory to meet its advertisers’ demand. Effectively integrating Google Ads with existing publisher-facing platforms would have benefited both Google Ads advertisers—by increasing their access to inventory—and Google—by increasing advertising sales, and in turn Google’s total revenues as a percentage of those sales. Instead, Google sought to maintain more control over advertising purchases made by its Google Ads’ advertisers. In particular, it limited the ability of its Google Ads’ advertisers to buy inventory from Google’s rivals. Google recognized that if it could secure access to its own pool of publisher inventory, it could control the entire transaction, end-to-end, and become the “the be-all, and end-all location for all ad serving.” To that end, Google built and launched its own publisher ad server, but the product failed to gain traction.

79. Rather than innovate and compete, Google found a shortcut. In 2007, Google announced that it would buy DoubleClick for \$3.1 billion. DoubleClick offered the industry-leading publisher ad server, called DoubleClick for Publishers or “DFP”, which at the time had an estimated 60% market share. DoubleClick also was developing a nascent ad exchange, called AdX.

80. The DoubleClick acquisition was a pivotal moment for Google’s display advertising technology business and its strategy to dominate the ad tech stack. The deal provided Google with direct access to website publishers (and their inventory) on DoubleClick’s publisher

ad server and, for the first time, a significant presence on both the advertiser and publisher sides of the ad tech stack. Google feared that if a rival acquired DoubleClick, Google would not control all the tools that link Google’s advertisers with publisher inventory; in short, a rival could “disintermediate” Google. Disintermediation risked allowing another company to control how and where publishers sell impressions to advertisers, something Google would not tolerate because it would limit Google’s ability to generate monopoly profits. Setting the stage for what was to come, the DoubleClick acquisition provided Google the unilateral power to implement a series of anticompetitive restraints, using its dominance on both the publisher and advertiser sides of the market to inhibit competition across the entire ad tech stack.

81. The Federal Trade Commission (“FTC”) investigated Google’s proposed acquisition of DoubleClick. The FTC considered “the possibility that Google could leverage DoubleClick’s leading position in third party ad serving to its advantage in the ad intermediation market” and whether Google could “exclusively bundle AdWords [advertiser demand] with [its publisher-side platforms] AdSense and DFP.”

82. The FTC ultimately declined to challenge Google’s acquisition of DoubleClick for the reasons set out in its public closing statement. The FTC concluded that “DoubleClick does not have market power despite its high market share”—over 60% at the time—and that “firms can and do switch ad serving firms when it is in their self-interest to do so.” Based on these assumptions, the FTC believed any anticompetitive conduct by Google “would likely be defeated by customers switching to one of the other third-party ad serving products.”

83. Google’s contemporaneous business documents paint a very different picture, however. Six months after the FTC closed its investigation without taking action, one senior Google executive wrote about the importance of controlling access to publisher inventory

through the DFP publisher ad server, stating “the thing we want ‘secured’ is the DFP platform adoption” because “if we have this” then Google would “have a +20% monetization advantage.” In other words, Google believed it could sustainably charge a higher price on advertising transactions than its competitors because it controlled the process and rules by which publisher inventory could be sold.

84. Google knew the emerging ad tech market better, and acknowledged in internal documents that “due to [the publisher ad server’s] position as the operating system for ad sales, switching costs are very high.” Because of this “stickiness” of publisher ad servers, Google knew it could manipulate the system in its favor—and to the detriment of competition—without fear that publishers would switch to other publisher ad servers. After the acquisition was completed, the former DoubleClick CEO explained at an internal Google strategy meeting, “My view is nothing really matters but the platform [publisher ad server]. Nothing has such high switching costs. If there’s a better network or exchange, you can just switch to it. Switching platforms is a nightmare. Takes an act of God to do it.”

85. Following the DoubleClick purchase, Google cemented its position as the dominant intermediary between advertisers and publishers through a series of additional acquisitions that eliminated potential competitors and further bolstered Google’s position in open digital advertising. For example, in 2009, Google paid \$750 million to purchase AdMob, a technology system that allowed publishers of mobile apps to sell ads as well. While Google’s conduct in the distinct market for mobile app advertising is outside the scope of this Complaint, Google’s anticompetitive conduct in the mobile apps market is consistent with the conduct alleged in the market for display advertising.

86. In 2010, Google acquired Invite Media for approximately \$81 million. Invite Media offered a demand side platform. Google subsumed Invite Media into a demand side platform it was developing, Display & Video 360 (at the time, known as “DoubleClick Bid Manager”). By capturing an increasingly large share of bigger, more sophisticated advertisers and advertising agencies, Display & Video 360 complemented Google Ads and expanded Google’s control over advertiser demand.

87. In 2011, Google bought AdMeld for approximately \$400 million. As discussed further below, AdMeld had developed technology to provide “yield management” functionality to publishers. Yield managers like AdMeld helped publishers manage inventory and optimize revenue by comparing offers from multiple advertiser demand sources at the same time. This comparison feature made it easier for new ad exchanges and advertiser demand sources to enter the ad tech industry because it gave publishers the incentive and ability to switch between ad exchanges and advertiser demand sources in response to better prices and service.

88. The DoubleClick, Invite Media, and AdMeld acquisitions helped Google achieve dominant positions at each level of the open web ad tech stack and set the stage for Google to control and manipulate the process by which publishers sell and advertisers buy open web display inventory.

B. Google Uses Its Acquisitions and Position Across the Ad Tech Stack to Lock Out Rivals and Control Each Key Ad Tech Tool

89. After amassing a position as the dominant intermediary for display advertising, Google used its monopoly power over each level of the ad tech stack to reinforce its dominant positions and limit where and how other ad tech providers could compete. Most notably, Google made its Google Ads’ demand available only through its AdX ad exchange. In turn, Google effectively made its ad exchange available only to publishers using its publisher ad server (DFP).

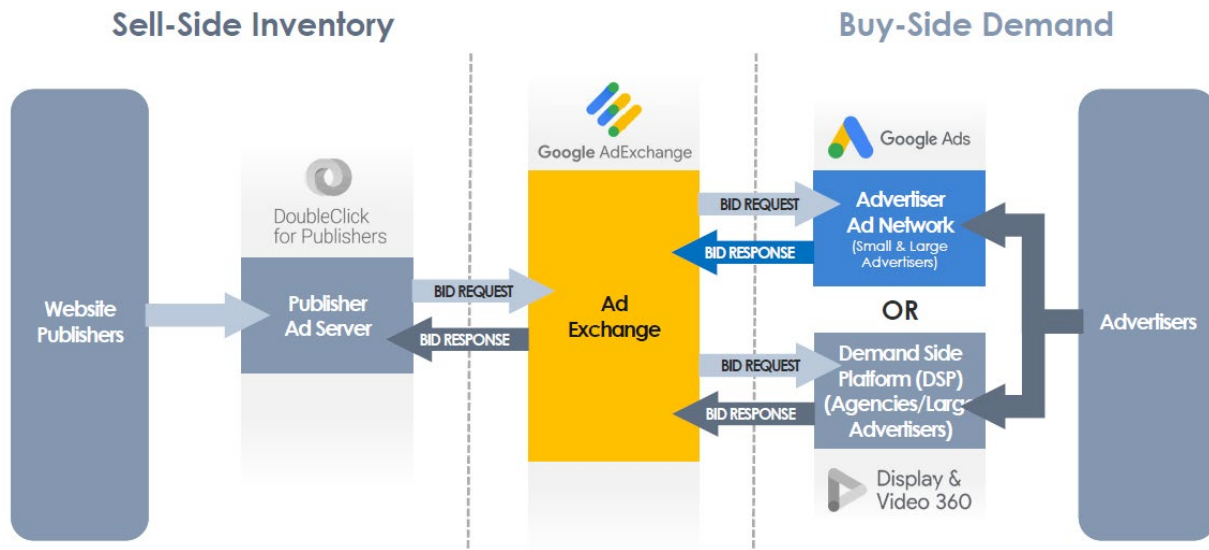
This benefited Google's long-run aspirations of dominating the publisher ad server market, at the expense of Google Ads' advertisers seeking access to the widest variety of publisher inventory at the lowest price. By allowing only its own publisher ad server effective access to important, unique Google Ads' demand, Google could force publishers to adopt and remain on its publisher ad server; other ad servers could not compete to offer a similar product. But this restriction meant Google Ads' advertisers could not buy inventory available only on other ad exchanges or via non-Google publisher ad servers, and they could not take advantage of free competition that might make that advertising inventory less expensive. The restriction also was contrary to Google's short-term financial interests, which turned on buying more and more-valuable advertisements from as many publishers as possible.

90. Google introduced several policies and auction changes to force more transactions to flow through its platforms and make it more difficult for publishers to switch ad servers. These restrictions collectively stifled competition by artificially preventing rivals from competing on the same terms as Google's products, thereby impeding publishers' and advertisers' ability to work effectively with rivals and allowing those rivals to obtain scale. Google implemented these restrictions even though it knew that it would have been better for Google Ads' advertisers (and far more profitable for Google Ads in the short run) to multi-home across ad exchanges. And, likewise, publishers would have benefited from being able to effectively access advertiser demand through multiple ad exchange intermediaries. These restrictions had the effect of taking Google's publisher ad server from a market leader to a monopoly—currently with no credible competition—and catapulting its nascent ad exchange into a monopoly position that dwarfs all other ad exchanges.

1. Google Thwarts Fair Competition by Making Its Google Ads' Advertiser Demand Exclusive to Its Own Ad Exchange, AdX

91. The DoubleClick acquisition was a first step in Google's march to monopoly. After purchasing DoubleClick, Google restricted Google Ads' purchasing of display inventory to sources controlled by Google (inventory owned by Google or by publishers using Google's monetization products, including its newly acquired publisher ad server). The goal was ultimately to lock publishers into its ad exchange and publisher ad server, and block competing ad exchanges and publisher ad servers from accessing Google's valuable pool of advertiser demand. Google implemented this restriction when it launched "AdX 2.0" on September 17, 2009. At the time, Google identified one of AdX's two differentiators from other ad exchanges as unique "access to AdWords advertisers."

92. Google has continued to sacrifice profits and act against the interests of its own advertisers by blocking its Google Ads' customers from buying almost any inventory through non-Google platforms, such as rival ad exchanges and networks, even if those competitors offered more valuable inventory or the same inventory at lower prices. Google estimated that by 2017, Google Ads was forgoing \$863 million per year in revenue by not purchasing inventory from rival ad exchanges and networks. But Google believed this forgone revenue was worth it in the name of advancing its growing moat and protecting its monopoly positions across the ad tech stack. Exclusive access to Google Ads' demand compelled most publishers to adopt whichever ad tech tools Google required to effectively access that demand.

Fig. 6⁹

93. Google Ads' unique and sizeable advertiser demand is what makes Google's ad exchange unavoidable for most website publishers. Google Ads' demand is unique both in its volume and diversity of advertisers (now with more than two million) and in its ability to attract advertisers at scale who cannot effectively use any other digital display advertising tool to purchase ads on the open web. Even for Google Ads' advertisers who can use alternative buying tools, many of them are pushed into Google Ads in order to buy other forms of critical advertising inventory that Google makes available effectively only through its buying tools, such as portions of YouTube, Gmail, and Search ad inventory. Together, these characteristics mean that Google Ads has a significant scale advantage by controlling a unique demand group (that spends about \$11.5 billion on display inventory each year).

94. Google Ads is also differentiated from other sources of advertising demand because Google's data-targeting advantages allow it to identify inventory that is uniquely

⁹ Diagrams are provided throughout to highlight the location within the ad tech stack where the conduct predominantly occurred. They are not intended to identify all areas where the identified conduct impacted the competitive process or other market participants.

valuable to Google Ads' advertisers. Google Ads' targeting data is derived from a wide array of user data that Google compiles across its many market-leading or monopoly products (e.g., Chrome, Gmail, Google Search) as well as data that Google requires its publishers to share with it through a data pool called the "ICM Coop."¹⁰ Google Ads combines this targeting data with contextual data Google extracts while crawling publisher websites. These sources of data fuel the immense network effects that raise barriers to entry and insulate Google from competitive pressure.

95. The advertiser make-up and data advantages of Google Ads lead it to buy large swaths of inventory that otherwise would go unsold. Certain inventory is valuable only to advertisers that use Google Ads exclusively; other inventory is undervalued without the user targeting and contextual data that Google makes available only to Google Ads. Google does not simply limit access to this data to its own advertiser buying tools. It also has exercised its market power to undercut rivals' ability to compete using the same or similar data. For example, after the DoubleClick acquisition, Google "hashed" (i.e., masked) the user identifiers that publishers previously were able to share with other ad technology providers to improve internet user identification and tracking, impeding their ability to identify the best matches between advertisers and publisher inventory in the same way that Google Ads can. Of course, any purported concern about user privacy was purely pretextual; Google was more than happy to exploit its users' privacy when it furthered its own economic interests.¹¹

¹⁰ The ICM Coop is not a real cooperative among website publishers. Rather, it is a data pool over which Google has sole control that publishers must participate in in order to receive competitive bids from Google Ads. Google estimated the value of this data to Google Ads was \$4 billion in 2015.

¹¹ See *supra*, n. 3.

96. Google recognized the power it wielded in Google Ads and used it to prop up and insulate its other ad tech products from competition. According to a 2011 internal analysis, Google found that allowing Google Ads to buy inventory on rival ad exchanges would cause serious losses to its publisher platforms (DFP and AdX) because many publishers and advertisers would prefer to transact through rival platforms. Google estimated that in such a scenario its ad exchange would lose 20 to 30% of its impressions, and its publisher ad server would lose 20% of its publishers. Later internal studies confirmed Google Ads' stranglehold. A 2014 Google experiment found that more than half of the impressions that publishers offered on its ad exchange would go unsold without the critical Google Ads' demand. If the Google Ads' demand was removed from the ad exchange, Google's publishers would experience a 65% drop in revenue because no advertisers outside of Google Ads were interested in buying the unique impressions available or able to do so in light of the auction restrictions described below. Google congratulated itself on having effectively locked out meaningful competition. At one Google strategy meeting, Google executives applauded the fact the "unique Google Display Ad demand" allowed it to justify "why we can charge 20%" fees for open auction transactions won on AdX, even on transactions that did not use Google Ads' buying tools.

97. By preventing publishers from accessing this incredibly valuable demand through rival ad exchanges that publishers otherwise would prefer, Google distorted the way in which website publishers partner with Google's competitors. As Google's former head of global strategy and commercialization explained: "When [advertiser] demand can only be found through certain sources, it compels publishers to work with that product." Because Google owns both DFP and Google Ads, and publishers needed to use DFP to access Google Ads' demand

effectively, Google had no incentive to operate DFP for its own publisher customers' benefit in the way that DoubleClick did.

98. For Google Ads' single-homing advertisers, Google has made it impossible for any rival ad exchanges to compete for the opportunity to link them with publisher inventory. As a result of this restraint, single-homing advertisers have more limited access to advertising inventory (less "reach"). Google recognizes that while this exclusivity locks in publishers, it harms Google Ads' advertisers, which have no reasonable alternatives to which they can turn. As explained in one 2012 internal document, the policy amounts to a "buyside-subsidizes-sellside model" that "artificially handicap[s] our buyside (GDN [Google Ads]) to boost the attractiveness of our sellside (AdX [ad exchange])." Later, in 2014, one Google employee complained about Google Ads' sending Google's publisher platforms a "\$3bn yearly check by overcharging our advertisers to ensure we're strong on the pub[lisher] side." These complaints by Google's own employees working on Google's advertiser tools reflect the artificiality of the restrictions imposed on Google Ads, and make clear that the restrictions are what they seem: blatant exclusionary conduct designed to obtain and maintain monopoly power rather than efforts to build a viable, vibrant ad exchange. Deliberately overcharging its own advertisers is also clear evidence of monopoly power over the advertising side of the ad tech industry. No other competitor could engage in such conduct and expect to stay in business.

99. In response to pressure from within Google's own ranks, including by employees managing Google Ads, and only after it shored up its market position across the ad tech stack, beginning in 2015, Google allowed limited categories of advertising demand to bid for some inventory on rival ad exchanges. This was something that Google's engineering team had previously considered and viewed as technically possible, but which its product leadership had

refused to allow. Google's buy-side employees championed this new feature as in the interest of Google Ads' advertisers and Google Ads itself, as a standalone product.

100. Even for the limited subset of demand that Google finally allowed Google Ads to bid on rival ad exchanges, Google disadvantaged those bids. Specifically, Google refused to allow Google Ads' advertisers to submit bids to rival exchanges using the same bid modelling and targeting data that Google Ads used to generate bids for Google's own ad exchange. Likewise, when submitting Google Ads' advertisers' bids to rival exchanges, Google submitted only the single highest bid, whereas when Google submitted the same advertisers' bids to Google's own ad exchange, AdX, Google submitted its two highest bids to improve the revenue payout to publishers.

101. In effect, Google systematically decreased the payout that Google Ads provided to publishers by extracting higher fees on the transactions (now ranging from 32% to 50%). In aggregate, Google understood "32% margin and no 2nd price makes [Google Ads demand] less desirable to access via a middle-man." Once again, Google acted to preserve its own monopoly power rather than its customers' best interests.

102. Google implemented these changes not because it was interested in helping its advertisers achieve the best return on their advertising investment; rather, by submitting two bids, Google was able to redirect sales back to AdX, ensuring AdX's "must have" status in the industry and making it difficult for rivals to have the scale necessary to compete. In fact, Google has gone so far as to enter into non-disclosure agreements with these rival ad exchanges to prohibit them from telling publishers that even this limited form of Google Ads' demand could be found outside of Google's ad exchange. Google imposed these restrictions because it understood that "when our competition is able to say that they have access to [Google Ads],

whether it is equal or not, our sales and marketing teams will have to be prepared for significant competitive pressure.” Rather than face that competitive pressure, Google simply restricted rival exchanges’ ability to market their capabilities to their publisher customers.

103. Google’s coupling of Google Ads’ demand to its budding ad exchange was significant in Google’s plan to dominate the market and exclude competition. Google took its existing scale advantages in Google Ads and extended those to Google’s other ad tech products, by driving more opportunities and transactions through them and away from rival marketplaces. As predicted, Google’s nascent ad exchange grew exponentially after its relaunch in 2009. In the wake of the DoubleClick acquisition and the implementation of these restrictive policies, Google’s ad exchange display revenue grew 283% in 2009 and an astounding 844% in 2010. By June 2011, Google executives boasted about becoming the “#1 player” in U.S. display advertising, a substantial jump for a company that had failed to gain traction with its own publisher ad server just five years earlier. Today, because of this exclusionary conduct, 95% of Google Ads’ spend flows through Google’s own AdX ad exchange, while less than 5% flows through rival ad exchanges. The combination of Google’s acquisitions with its anticompetitive business practices has suppressed, or altogether eliminated, the necessary growth for rivals to compete.

2. *In Turn, Google Makes Its Ad Exchange’s Real-Time Bids Exclusive to Its Publisher Ad Server*

104. At the same time, Google used what most publishers saw as AdX’s must-have status to reinforce and grow its already dominant publisher ad server, DFP, ultimately pushing remaining publishers to adopt Google’s ad server and forcing rivals to exit the market. With its relaunch of the ad exchange after the DoubleClick acquisition, Google required publishers to use its ad server to obtain real-time bids from its ad exchange. If a publisher chose not to use DFP, it

was relegated to selling impressions to AdX at a floor price based on historical average prices, which were often much less than the real-time, impression-specific bids AdX made through DFP. As a 2018 Google presentation bluntly noted, “[DFP] is the only way to access [AdX] as a publisher.”

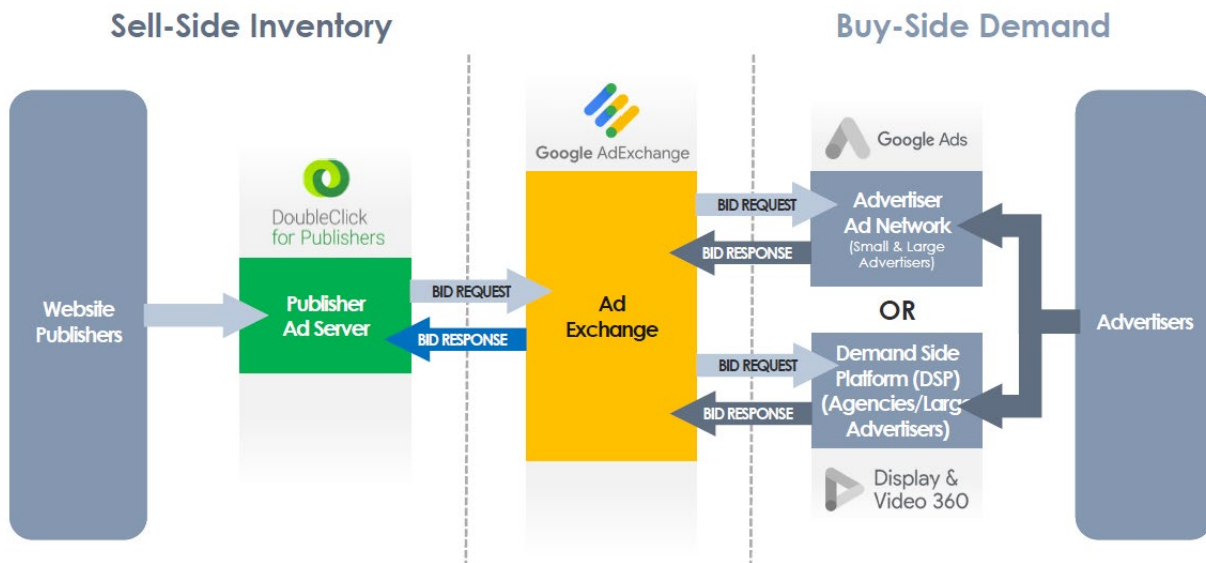


Fig. 7

105. Google did not need to make AdX exclusive to DFP in this way, but nevertheless decided to pursue the most restrictive alternative. In 2011 and 2012, Google developed a feature that would enable AdX to compete for inventory in the same way on other publisher ad servers. The feature was available in beta to some partners and required only “minimal effort” to roll out commercially. But Google saw that this feature risked taking away a “key differentiator for DFP”: access to real-time AdX demand. As one Google employee explained in September 2012, “it is too early to give AdX to non-XFP [DFP] partners. . . . This is an amazing time to ‘lock in’ impressions by offering XFP [DFP] to publishers AdX can serve as a tool to pull publishers onto XFP [DFP]. . . . Ad Servers are sticky, and hard to replace. The next 12 months are a very good time to switch publishers over.” By 2013, Google decided to end this experiment in

openness, with Google's lead product manager for AdX declaring: "[O]ur goal should be all or nothing – use AdX as your SSP or don't get access to our demand."

106. By limiting Google Ads' demand to the AdX ad exchange, and limiting real-time access to the ad exchange to publishers using Google's publisher ad server, Google compelled publishers to adopt its ad server for effective access to Google Ads' demand. Recognizing the importance of real-time competition for ad inventory—which priced an advertisement based on the particular characteristics of a webpage user at that specific point in time—Google blocked publishers using rival ad servers from connecting to Google's ad exchange in the same way. Google knew that its Google Ads' advertisers provided a unique, rich source of advertiser demand, and that no other publisher ad server (or ad exchange) could offer similar access to such a lucrative pool of advertiser demand. Many publishers could not afford to use a rival publisher ad server because they could not afford to lose the revenue that Google's exclusively-linked platforms could provide. In essence, Google dictated publishers' choice of each key ad tech tool used to sell their inventory: publishers must make their inventory available through Google's publisher ad server and ad exchange to get the opportunity to sell a portion of it to Google's extremely valuable Google Ads' advertisers.

107. By 2015, these restrictions had virtually eliminated competition between publisher ad servers, driving rivals to abandon the market completely. Google's market share for publisher ad servers soared from 60% in 2008 to 90% by 2015. In a 2016 customer presentation, Google described DFP as the "defacto [sic]" publisher ad server with a "90% market share." At the same time, it guaranteed to Google Ads—via Google's ad exchange—preferential access to an unrivaled swath of publisher inventory, as well as the associated contextual and user targeting data, which supported Google Ads' remaining a dominant, scaled ad network.

108. To this day, Google’s restrictions and conduct essentially foreclose the possibility of entry into the publisher ad server market. To enter the publisher ad server market, a competitor not only would need to offer a full-featured ad server, but also would need to be able to link it to an ad exchange and advertiser demand source of the same size and scale as Google’s ad exchange and Google Ads’ advertiser demand. Without such a full-stack offering, a competitor would need to convince publishers to sacrifice effective access both to Google’s ad exchange as well as access to Google Ads’ unique advertiser demand, an ask that is simply not economically feasible given Google’s successful exclusionary conduct to date.

3. *Finally, Google Uses Its Control of Publisher Inventory to Force More Valuable Transactions Through Its Ad Exchange*

109. Google’s ownership of the leading publisher ad server, DFP, allowed it to set the rules that governed how most publisher inventory in the market is sold. Google internally referred to publisher ad servers as the “ad revenue operating system for publishers” because they decide who is offered a chance to buy publisher inventory and on what terms. Not content to operate in a free and competitive market, Google altered its publisher ad server rules to force more transactions—and more high-value transactions—through its ad exchange and advertiser platforms. The changes did not allow Google’s ad exchange rivals to compete in the same way or

on the same terms, largely leaving them with the leftover scraps of inventory that Google's advertisers did not want, even at artificially discounted prices.

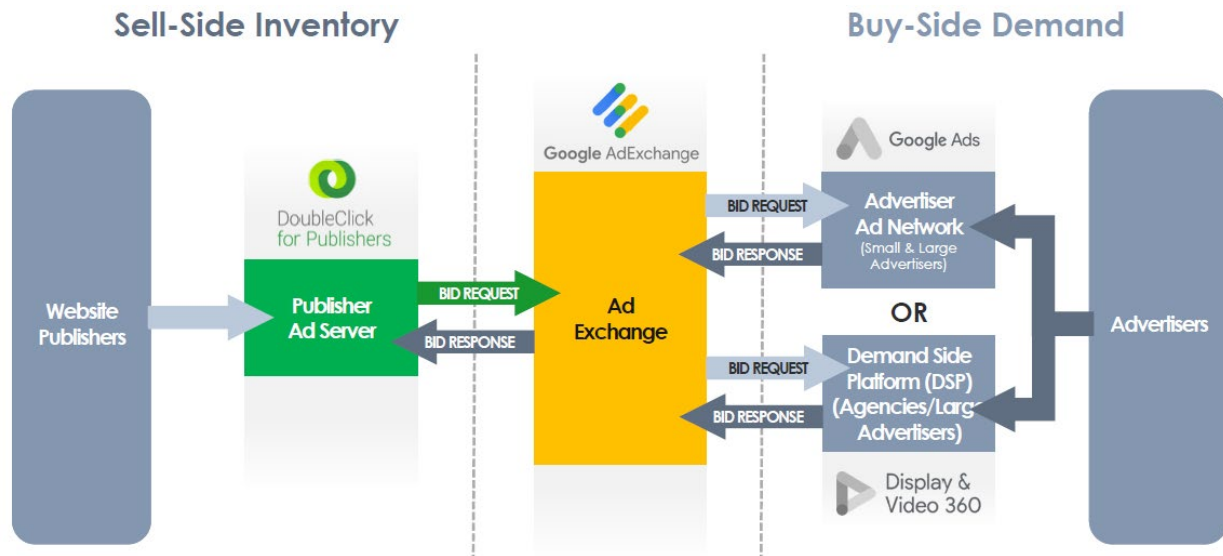


Fig. 8

110. Until at least the advent of header bidding between 2012 and 2013 (and for many publishers not until at least 2018), publishers that wanted to offer inventory to multiple ad exchanges via Google's publisher ad server had to use a system known as the "waterfall." Even though this system plays a smaller role now than it once did,¹² it played a pivotal role in establishing Google's dominance in the ad exchange market, was a critical predicate to certain other Google conduct, and helped to create the market monopoly that Google enjoys today.

111. Under the waterfall process, the publisher ad server would send offers to sell advertising inventory to ad exchanges and advertiser ad networks one at a time in sequence until it found an eligible buyer. To set up the waterfall, publishers had to manually enter into the publisher ad server the average price they expected to be paid by each ad exchange based on

¹² Because of the difficulties and costs of utilizing newer alternative systems in Google's ad server, many publishers are still forced to use the waterfall system today.

historical averages. Because these were average prices, they did not necessarily reflect what an ad exchange would pay for any individual impression at any particular time. The publisher ad server then ranked each ad exchange from highest to lowest based on average historical price. Then when a user opened a publisher's webpage and an ad impression became available for sale, the ad server offered the impression to the ad exchange ranked highest in the waterfall. If that ad exchange had an advertiser willing to pay more for the impression than the minimum price set by the publisher (the "price floor")—which could differ from the average prices of that ad exchange—the ad exchange won the impression, and its advertiser was able to display the ad. The ad was not submitted to any of the other ad exchanges in the waterfall, even if one of them might have been willing to pay more for the impression. Alternatively, if the first ad exchange did not have an advertiser willing to pay at least the publisher's price floor, the ad server called the next ad exchange in the list. This process continued until someone purchased the impression or the last ad exchange in the waterfall was called.

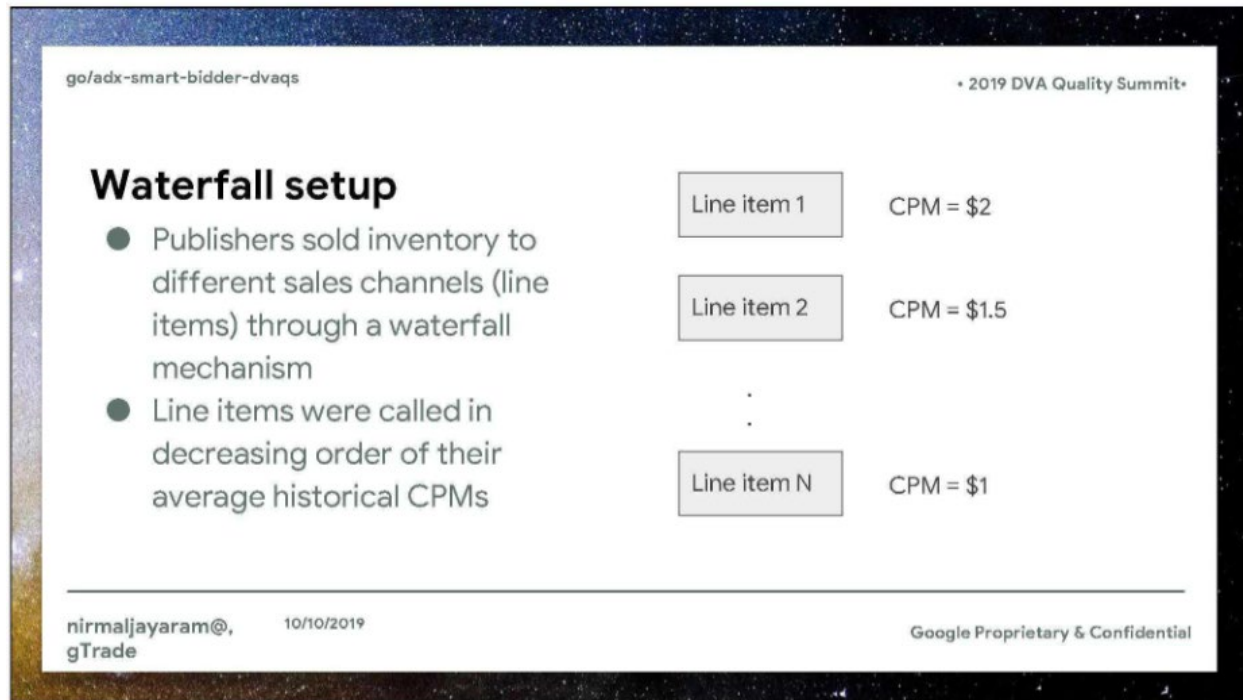


Fig. 9

112. The inefficiencies associated with the waterfall system are obvious: ad exchanges at the bottom of the waterfall might never get a chance to bid, even if they could supply a lucrative bid. In those instances, publishers received less revenue than they could have. But while this inefficiency plagued how inventory was sold to rival ad exchanges, Google used its control over the process to allow its ad exchange—and only its ad exchange—to compete outside of the waterfall process.

113. As part of its post-acquisition relaunch of AdX, on a “system written from scratch” on Google’s platform, Google redeployed “dynamic allocation.” Dynamic allocation provided AdX a prized position over all other indirect sources of advertising demand, which allowed AdX to both “see more” and “win more” valuable publisher inventory.

114. *First*, Google configured its publisher ad server to afford Google’s ad exchange a “first look” at all inventory the ad exchange was eligible to buy. Google’s publisher ad server

always called Google's ad exchange for a real-time bid before offering inventory to rival ad exchanges. This placed Google's ad exchange at the top of every waterfall, regardless of where it would otherwise be ranked based on its average historical prices. In practice, it meant that Google's ad exchange saw more publisher inventory than any other ad exchange and could offer advertisers the ability to obtain the most valuable impressions by simply paying slightly more than a static historical *average* price paid by rival ad exchanges.

115. *Second*, before Google's ad exchange competed for an impression, Google's publisher ad server shared with its ad exchange the highest competing price from the waterfall, i.e., the highest *average* price of a rival ad exchange. This set the auction floor price within Google's ad exchange and provided bidders on Google's ad exchange with two key advantages: (1) buyers on Google's ad exchange could see the floor price (i.e., the minimum price to win) and adjust their bids accordingly; and (2) buyers on Google's ad exchange often had to pay only that *average* price of the rival ad exchange. The latter of these advantages was a function of Google conducting a second-price auction on its ad exchange. Under this auction format, if only one bid on Google's ad exchange was higher than the price floor, that bid won the inventory at the floor price that had been set by the rival ad exchange's average price. In this way, Google's ad exchange was able to win high-value impressions without paying the price advertisers on other ad exchanges were actually willing to pay.

116. *Third*, Google configured the ad server to allow its ad exchange to compete on the basis of real-time pricing derived from its internal auction for a particular impression shown to a particular internet user. Unlike rival ad exchanges, Google's ad exchange was not relegated to competing on the basis of historical average prices. Combined with Google's treasure trove of user targeting and webpage contextual data, Google's control over the ad server allowed it to

tailor its bids more carefully; that is, it could bid high for a more valuable impression and low for a less valuable impression. For example, it could offer a publisher \$10 CPM to show a car dealer's advertisement to a user who recently clicked through several car manufacturer websites while offering the same publisher only \$1 CPM to show the same ad to a 14-year-old user who resides in a state where the dealer does not operate. Google's publisher ad server would not permit other ad exchanges to compete in this way. Instead, all other ad exchanges were forced to compete on the basis of the "waterfall" method using historical, average prices, even though the industry quickly developed a technology standard to bid in real time in this way.

117. This two-tiered arrangement denied rival ad exchanges the opportunity to gain the scale needed to compete effectively with Google by diverting bidding opportunities and transactions to Google's ad exchange and away from rivals who did not have a chance to compete at all or to compete on the same terms. It also harmed publishers in the form of lower revenues, limited the ability of advertisers to identify publisher inventory they valued most at the best prices, and decreased the overall quality of matches between publishers and advertisers.

118. Under the waterfall setup, rival ad exchanges never had the opportunity to bid on most impressions. If an ad exchange earlier in the waterfall sequence submitted a bid above the publisher's price floor, the ad server never offered the inventory to ad exchanges lower in the waterfall. The rules that Google's publisher ad server applied to Google's ad exchange, however, provided Google's ad exchange the opportunity to bid on every eligible impression, armed with substantial data on the publisher's inventory and the competitive landscape. Because rival ad exchanges were relegated to the waterfall process, unlike Google's ad exchange, they had limited windows into the universe of publisher inventory available and lacked the valuable data on available inventory and competition that Google harvested. By preventing publishers from freely

multi-homing and seeing real-time bids from multiple ad exchanges, Google deprived publishers of the benefits of full competition between ad exchanges. Likewise, by providing Google's ad exchange with a preferential—and for many impressions, sole—opportunity to buy publisher inventory, Google discouraged advertisers from multi-homing among ad exchanges and provided a substantial competitive advantage to buyers on Google's ad exchange, the largest buyer being Google Ads.

119. In addition, through dynamic allocation, Google's ad exchange had the opportunity to win impressions whenever it matched a rival's average price. This permitted Google's ad exchange (and its largest buyer, Google Ads) to win more impressions than its rivals, especially higher-value impressions. But for dynamic allocation, a rival ad exchange might have won the impression because it could offer a higher price or better match. Over time, this distortion of the auction process meant that advertisers were more likely to win the impressions they most wanted through Google's ad exchange as compared to a rival ad exchange. As a result, rival exchanges struggled to attract advertiser ad campaigns, which in turn made it difficult for them to amass publishers willing to offer their inventory through the ad exchange. Of course, dynamic allocation also hurt Google's own publishers, by sacrificing the fees they paid Google to maximize the value of their advertising inventory.

120. In 2014, Google expanded and further entrenched its artificial advantages by introducing “enhanced” dynamic allocation, which remains in place today. This update allowed Google's ad exchange to obtain the benefits of dynamic allocation over inventory potentially covered by direct contracts between publishers and advertisers. Historically, this inventory was not offered to ad exchanges at all because qualifying inventory was set aside to fill the direct contract; only after the direct contract was filled did otherwise qualifying inventory become

available for auction. Enhanced dynamic allocation afforded Google’s ad exchange a right of first refusal over this inventory regardless of whether the publisher had yet fulfilled the terms of the direct contract. Enhanced dynamic allocation allowed Google’s ad exchange to win the impression as long as it was willing to pay more than Google’s own estimate of the “value” of fulfilling the terms of the direct contract at that moment, which Google calculated through an opaque process that predicted the likelihood the publisher would still be able to satisfy the terms of the direct contract through future impressions even if Google’s exchange filled the one currently available. At the same time, Google ensured that “[i]t [was] not possible for publishers . . . to deactivate Enhanced Dynamic Allocation” within the publisher ad server.

121. Combined, dynamic allocation and enhanced dynamic allocation push more transactions through Google’s ad exchange by unfairly tilting the playing field in Google’s favor, driving additional scale benefits available only to Google. Because of the exclusive link between Google’s ad exchange and its market-leading publisher ad server, no rival can offer publishers or advertisers the same terms as Google. The benefits to Google, and only Google, are plain.

122. *First*, Google has been able to apply its substantial 20% revenue share fee at the ad exchange level over more transactions, boosting Google’s revenues and profits. This fee has been earned not only on transactions where Google Ads won, but also on transactions where other Google and non-Google advertiser buying tools won. Because Google could capture these higher revenues at the ad exchange level, Google was able to forgo or heavily discount the fees it otherwise might charge for publisher ad server services—historically much smaller than ad exchange fees. Indeed, for many customers, Google completely waived publisher ad server fees on a given transaction if it was able to charge its 20% ad exchange fee. By extracting higher fees at the ad exchange level than at the publisher ad server level—which Google needed to control to

force more transactions to its ad exchange—Google has also able to maintain low ad serving fees while still achieving its margin goals across the ad tech stack. This fee structure discourages entry by potential ad server competitors, because entry could only be economically feasible by replicating Google’s overall strategy: building dominant positions at each level of the ad tech stack and forcing more transactions to flow through those tools.

123. *Second*, by forcing more transactions through Google’s ad exchange and away from rivals, Google has distorted the pathways through which publishers and advertisers transact and impeded the ability of competitors to gain the scale necessary to compete effectively in the ad exchange market. Google’s dynamic allocation and enhanced dynamic allocation programs have decreased the likelihood that a rival ad exchange could win a transaction, even if it had an advertiser willing to pay the most for an impression. In turn, this has diminished the ability of ad exchanges to attract additional publishers and advertisers to their platforms and has deprived them of valuable transaction data that could improve their competitiveness.

124. *Third*, by giving Google’s ad exchange (and only Google’s ad exchange) a “first look” option of purchasing publisher impressions offered for sale through DFP, Google has limited the ability of publishers to freely and effectively offer their impressions for sale on multiple ad exchanges. Dynamic allocation and enhanced dynamic allocation has resulted in a two-tiered system—a special auction where Google’s AdX competed and a secondary, inferior auction potentially available to rival exchanges. Publishers are unable to partner with Google’s rival ad exchanges on the same terms as Google’s AdX. Those rivals cannot integrate with DFP via a mechanism equivalent to dynamic allocation, even if they had the technological capability to do so.

125. Even though Google modified the way dynamic allocation operated in late 2019, the effects of the decade-long program persist. Over that period, Google has amassed substantial scale within its ad exchange while undercutting rivals' ability to do the same. The flywheel network effects of that scale continue to advantage Google's ad exchange, especially when combined with the new algorithmic bidding programs described below that largely replicated the effect of dynamic allocation. Even today, Google continues to use enhanced dynamic allocation to favor buyers transacting through Google's platforms. Only those buyers can bid with knowledge of the Google-determined price floor that Google sets through enhanced dynamic allocation.

4. *Google's Dominance Across the Ad Tech Stack Gives It the Unique Ability to Manipulate Auctions to Protect Its Position, Hinder Rivals, and Work Against Its Own Customers' Interests*

126. In addition to restricting vital Google Ads' demand to website publishers using Google's publisher ad server and ad exchange, Google realized it also could manipulate Google Ads' bidding strategy to further entrench its publisher ad server and make entry by competing ad servers unworkable. Google Ads ostensibly bought inventory on behalf of its advertisers using price and budget limitations decided by each advertiser. But Google chose to do so in ways that served Google's long-term goal of dominating publisher platforms.

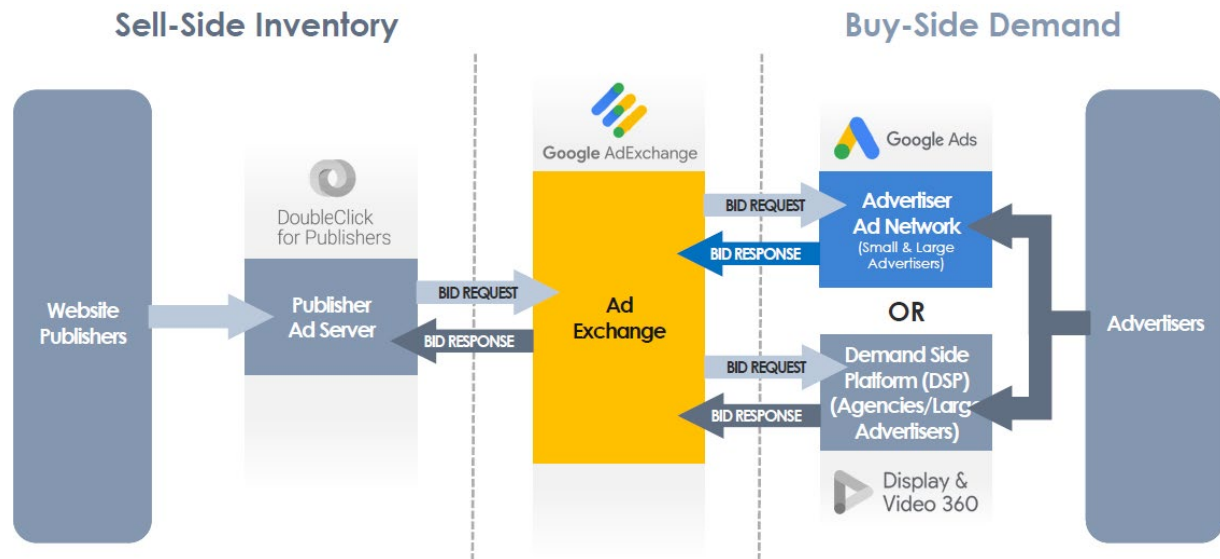


Fig. 10

127. Google Ads’ advertisers set maximum prices,¹³ budgets, and other parameters for their campaigns, but Google Ads is otherwise a “black box” to advertisers. Google has nearly full control over when, where, and how Google Ads bids for its advertiser customers. Using that control, Google designed a system intended to force Google Ads’ two million advertisers to pay higher advertising prices. The goal and effect are clear: increase payouts to publishers using Google’s platforms—the only place Google Ads’ demand was available—to make Google Ads’ advertising demand and Google’s publisher ad server and ad exchange even more indispensable to publishers (while also allowing Google to maintain its supra-competitive take rates). In doing so, Google foreclosed the ability of rivals to compete effectively against its publisher ad server business and further propelled Google’s DFP ad server from a dominant platform to a monopoly.

128. Over time, as Google’s monopoly over the publisher ad server was secured, Google surreptitiously manipulated its Google Ads’ bids to ensure it won more high-value ad

¹³ Although advertisers set a maximum price for advertising (generally on a per click basis), Google actually charges advertisers the lower of 1) their maximum price or 2) Google’s cost plus a set margin.

inventory on Google’s ad exchange while maintaining its own profit margins by charging much higher fees on inventory that it expected to be less competitive. In doing so, Google was able to keep both categories of inventory out of the hands of rivals by competing in ways that rivals without similar dominant positions could not. In doing so, Google preserved its own profits across the ad tech stack, to the detriment of publishers. Once again, Google engaged in overt monopoly behavior by grabbing publisher revenue and keeping it for itself. Google called this plan “Project Bernanke.”

a) Google Works Against the Interests of Its Google Ads’ Customers By Submitting Two Bids Into AdX Auctions

129. Google Ads determines how to bid on behalf of its advertisers using price and budget maximums decided by the advertiser. For each piece of available inventory, Google Ads runs an internal auction of Google Ads’ advertisers, based on an algorithm that considers eligible advertisers’ specified maximum cost-per-click prices, Google’s predictions of the likelihood of a user clicking an ad, and a number of other factors. These bids are then converted into a cost-per-impression (“CPM”) bid, which, until 2013, Google then adjusted downward to ensure Google Ads would charge an expected 14% take rate on each impression—in addition to the 20% take rate charged by Google’s ad exchange. If Google Ads then won the impression—and the user ultimately clicked on the ad—the Google Ads’ advertiser would pay the amount paid for the impression plus Google’s fee.

130. Until late 2019, AdX operated what was known as a second-price auction. In a second-price auction, the advertiser with the highest bid wins the right to display its ad on the publisher website. The winning advertiser, however, only pays one cent more than the price of the second-highest bid, and gets to keep the difference between the two as an “auction discount.”

If the advertiser is the only bidder, then it must pay only the minimum price the publisher agreed to accept, known as the price floor.

131. Because of this auction dynamic, advertiser buying tools have an incentive to submit only one bid into an ad exchange's auction. If an advertiser buying tool submits two bids into a second-price auction, the higher bid might win the auction while the lower bid sets the price; without the second bid, the inventory might have been sold to the winning advertiser at a lower cost (possibly even at the floor price set by the publisher). In essence, the second bid into the auction only serves to drive the final auction price upwards, while conveying no real increased chance of winning.

132. To avoid driving up the cost of advertising, non-Google advertiser buying tools only submitted a single bid into AdX auctions. But Google took a different approach for Google Ads and, unknown to advertisers, submitted *two* bids from Google Ads' advertisers into the AdX auction. This was contrary to the interests of Google Ads' advertisers—who benefited when Google Ads paid less for impressions resulting in clicks—but furthered Google's goal of locking publishers into its ad exchange and publisher ad server. These higher payouts for publishers on Google's platforms were a key part of Google's overall strategy to prevent new publisher ad servers from entering the market and to increase the stickiness of Google's own publisher ad server by raising publishers' switching costs. The strategy allowed Google to extract additional margins across the ad tech stack through the two-tiered auction structure described above. A publisher that left Google's platform not only lost access to all of the unique advertiser demand available only on Google Ads, but also lost access to an advertising buying tool willing to overcharge its advertisers to benefit its publishers.

133. Google generally did not disclose to Google Ads' advertisers the fees that Google extracted from their ad purchases. Even when it did publicly disclose average fees from the perspective of the revenue received by publishers, it did not disclose that those fees came on top of advertising prices that were inflated by virtue of Google Ads' bidding practices. This obscured the total "take rate" Google was keeping for itself, making it difficult for advertisers to compare Google Ads to any potential competitor.

134. Google's internal analyses confirm the purpose and effect of Google Ads' double-bid policy. A 2013 study found that Google Ads submitted the top two bids in **85%** of the auctions it won, meaning its bids set the price in the vast majority of auctions it won. Because Google Ads did not face meaningful competition for its advertising customers, however, this statistic did not lead Google to reduce its advertisers' bids or otherwise adjust its bidding strategy. Rather, the study confirmed Google's understanding that it had full control and pricing power over a unique pool of advertiser demand that was often interested in inventory other advertisers did not or could not value in the same way. As Google itself acknowledged, Google Ads had "no margin or inventory sourcing constraints" so it was able to "establish[] processes to tune margins in the backend," outside auctions. As usual, what mattered most was Google's own dominance, not its customers' best interests.

135. Years later, little had changed when Google revisited the question of what would happen if Google Ads submitted only one bid into the AdX auction.¹⁴ The answer: the 30 to 40% boost in publisher revenue from Google's two-bid strategy would disappear. Strikingly, the same analysis showed that if Google Ads submitted only one bid on AdX—and thereby was able to

¹⁴ On the limited occasions where Google Ads bids on inventory on third-party exchanges, it submits only one bid.

buy inventory at lower prices—its profits on Google Ads would increase by an astonishing 50% in the short run.

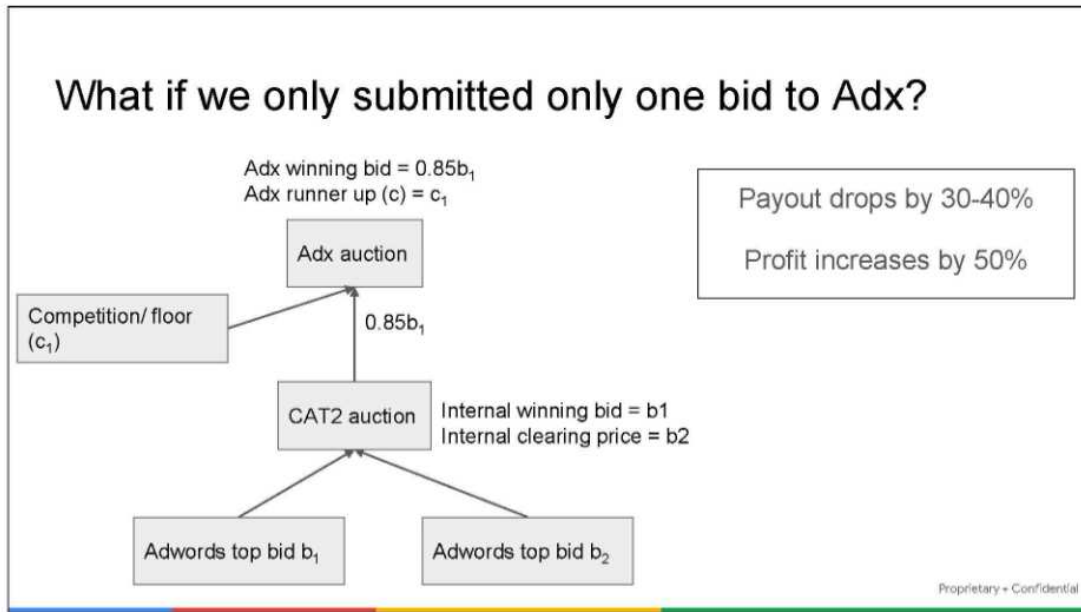


Fig. 11

136. But Google simply could not risk a change that weakened its ability to keep publishers locked into its publisher ad server and ad exchange. The loss of inflated publisher revenues on Google's platform might finally make a rival publisher ad server an attractive alternative and threaten Google's monopoly. If publishers switched, Google would lose control over the ad selection process and be forced to interoperate with those rival products for inventory.

b) Google Manipulates Its Fees to Keep More High-Value Impressions Out of the Hands of Rivals

137. As Google's publisher ad server monopoly was being cemented, Google's focus shifted to ensuring its ad exchange rebuffed growing challenges from rival ad exchanges. Recognizing that Google Ads still faced little competition for most impressions it won on AdX, over time Google adjusted its fees—and in turn its bids—to ensure it could win more high-value

transactions while paying less for lower-value transactions. By owning both the dominant publisher ad server and the dominant ad exchange, Google had unique access to the price data it needed to make these adjustments in a way that ensured more transactions, revenue, and profits flowed to Google—and in particular its Google Ads’ ad network and ad exchange—with minimal risk to Google.

138. First, in January 2013, through a program called Dynamic Revenue Share, Google adjusted the way Google Ads took its revenue share fee from a fixed 14% take rate on each impression sold to a changing, or dynamic, fee that averaged 14% per publisher over time. This allowed Google Ads to effectively increase its bids in competitive auctions (by taking a lower expected fee) and make up the losses by setting a higher expected fee on non-competitive auctions. This change reinforced Google’s ability to win more transactions on its ad exchange than could rival ad networks or demand-side platforms, augmenting the advantages Google already afforded its ad exchange through dynamic allocation, without the need to compete by reducing its fees.

139. Second, later in 2013, Google implemented Project Bernanke,¹⁵ which doubled-down on Dynamic Revenue Share by subsidizing bids (i.e., bidding above the advertiser’s willingness to pay) on competitive impressions, thereby sacrificing any profit on the transaction. Of course, Google ensured that its own margins would be maintained. Google offset any loss on a given transaction by charging much higher fees (i.e., 50% or more) on impressions where Google Ads faced no competition—the majority of impressions Google Ads had already been winning. In doing so, Google Ads and AdX were able to win more impressions over their

¹⁵ Project Bernanke was named after former Federal Reserve Chairman Ben Bernanke because it resembled “quantitative easing on the Ad Exchange.”

respective rivals, increasing Google Ads' spend by 20% and profits by 30%, and increasing overall ad exchange revenue by 8%. A Google simulation of the program confirmed that advertisers using non-Google buying tools won fewer of the coveted high-value impressions, decreasing their relatively smaller spend on AdX by 14%.

140. Finally, in 2014, Google implemented Project "Global Bernanke" which changed the method by which Google calculated the Google Ads' take rate (sometimes referred to as "margin"). Instead of applying the same take rate to each publisher's ad inventory, Google took an average take rate at the ad exchange level. Google took a higher cut of advertiser spend for some publishers while taking less for others. The effect was to further shift the publisher benefits of Google Ads' two-bid system to the most important publishers and away from "non-competitive" publishers (i.e., publishers whom Google believed were unlikely to risk switching to a rival ad server). Google candidly acknowledged that by 2014 it was not worried it might lose "non-competitive publishers." As one document explained, it is "unlikely they can do better on another network (which doesn't have any [Google Ads] demand)."

141. The Google-generated graphic below shows Dynamic Revenue Share and Project Bernanke in practice. After running its internal auction (as described above), Google Ads calculates its two highest bids on a CPM basis as \$1.00 and \$0.96 (the gray bars). These bids might be similar because they are based on the same Google targeting data. Applying a uniform 14% take rate (or "margin") would result in bids equal to \$0.86 CPM and \$0.83 CPM. With dynamic revenue share, Google adjusted the bids to \$0.95 CPM and \$0.83 CPM (the red bars in the Figure). For Bernanke, Google raised the first bid even further (sometimes substantially), as the first bid determines the winner of the auction. By raising the first bid (here from \$0.95 CPM

to \$1.20 CPM, the green bars), Google Ads won more auctions, either clearing publishers' reserve price more often or winning against a rival's bid for competitive impressions.

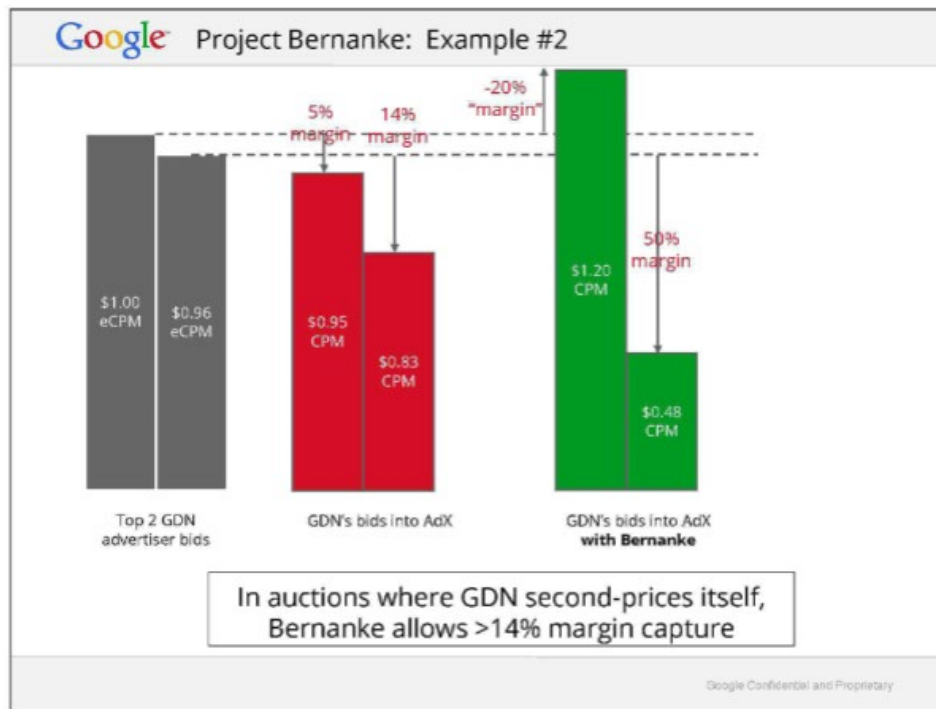


Fig. 12

142. By manipulating the auctions in this fashion, Google was able to subsidize the inflated advertiser bids by dropping the price of the runner-up's bid (here from \$0.83 CPM to \$0.48 CPM). Where an auction was not competitive—the majority of auctions that Google Ads won—the lower price was the one Google Ads paid for the impression. Google then kept the margin (an estimated 50% in the example) to subsidize competitive queries. In this example, instead of the website publisher receiving \$0.83 CPM for the advertisement, it received only \$0.48 CPM for the impression under Bernanke, assuming only Google Ads' advertisers submitted bids. A similar drop in price would occur for other "non-competitive" impressions. At the advertiser level, Google aimed for the same average take rate for each Google Ads' advertiser: a 32% difference between what the advertiser paid to Google and what Google ended

up paying to publishers for all of the impressions that the advertiser purchased.¹⁶ Bernanke increased the number of transactions that Google won through its platforms, and in turn increased Google's overall revenues and profits while denying scale to competing ad exchanges.

143. Project Bernanke allowed Google Ads to continue to pass along additional revenue to publishers on Google's platforms (the only place it ran) but did so disproportionately relative to the competitiveness of the publisher. In doing so, it won more high-value impressions on Google's ad exchange and reinforced the stickiness of Google's ad server for key publishers. Other ad exchanges and ad servers that lacked a captive source of advertiser demand whose bids they could manipulate were unable to subsidize important publishers in the same way, presenting another roadblock to entry or expansion in the publisher ad server and ad exchange markets. For their part, while Google Ads' advertisers won some additional competitive impressions, they did not receive the full benefit of the lower prices Google Ads paid for non-competitive inventory. Moreover, Google did not disclose to advertisers that it was shifting savings away from them to increase its own margins. In effect, Google fended off competition that could have challenged its monopoly power to force advertising transactions through its own ad tech products and limited publishers' and advertisers' ability to multi-home with rival products while still being able to maintain its high overall margins. In essence, a win-win, but only for Google.

144. In terms of its impact on competition, Google's dynamic allocation and dynamic revenue sharing programs functionally made price competition among rival ad exchanges obsolete because no rival had sufficient scale across the ad tech stack to compete against Google.

¹⁶ Google Ads' advertisers specify a maximum cost-per-click they are willing to pay. Google charges advertisers a fee on top of the price that Google pays for publisher inventory. The result is often less, and "sometimes much less," than the advertiser's specified maximum.

Google could effectively afford to charge nothing where it wanted to obtain high-value inventory because it had the ability to make up the difference on the back end with less valuable inventory.

C. Google Buys and Kills a Burgeoning Competitor and Then Tightens the Screws

145. By 2010, Google’s rivals had begun to design and market technology in an attempt to circumvent the limitations on competition that Google imposed through its publisher ad server. Google’s response was forceful, extinguishing the leading innovator via acquisition and tightening its restrictions to head off similar potential threats. Google continued to impede its rivals’ ability to offer real-time competition for publisher inventory on the same playing field as Google’s ad exchange. Google used anticompetitive means to keep customers on both sides of the stack (i.e., publishers and advertisers) locked in to its ad tech tools, while ensuring that competitors for those valuable customers were locked out.

1. Google Extinguishes AdMeld’s Potential Threat

146. In 2011, Google acquired a competitor, AdMeld. In doing so, Google removed from the market what it viewed as a “critical threat” to its ad exchange and publisher ad server businesses. AdMeld’s yield management technology could receive bids in real time from multiple ad exchanges and other demand sources. This could allow other ad exchanges to compete in the same way Google’s publisher ad server allowed Google’s ad exchange to compete through dynamic allocation, utilizing in part a real-time bidding standard. Publishers quickly moved to adopt the yield management technology because it allowed them to multi-home more effectively among ad exchanges and ad networks. It also gave publishers the ability to connect with the advertisers who especially valued their inventory. In contrast, Google’s publisher ad server did not permit connections to any advertising demand source other than the buyers on the AdX ad exchange, which of course included Google Ads’ advertiser base. Quite simply, AdMeld threatened to destroy the advantage Google had created for itself in its

exclusionary publisher ad server by allowing website creators to offer their digital advertising inventory to multiple ad exchanges in order to find the best available match.

147. Externally, AdMeld described itself as “the largest, independent practitioner of RTB [Real Time Bidding] behind Google,” “connect[ing] to more than 200 ad networks, & 35 Demands Side Platforms (DSP) and process[ing] more than 11 billion bids daily” with links to 20 leading data providers. Relying on its “core functionality” of real-time bidding, AdMeld proposed becoming its customers’ “central ad decision hub,” the key role Google reserved for its own publisher ad server product.

148. In a 2010 strategy discussion, Google executives noted that “Yield Managers are a threat we need to take very seriously” with “AdMeld [being] the largest concern” and one of three “Key competitors.” Specifically, if AdMeld continued to attract publishers to its technology, Google worried about the possibility of having to “pass real-time AdX pricing into a non-DFP ad server.” If Google were forced to do so, it would eliminate DFP’s exclusive access to AdX, which Google believed would be—and which ultimately was—the key to DFP’s growth and enduring dominance.

149. AdMeld typically charged only a 7% revenue share compared to Google’s 20% revenue share on AdX. So rather than compete with AdMeld, what did Google do? It bought and buried it. In a presentation outlining the “Strategic Rationale” for the deal, Google executives explained that the acquisition would “reduce [the] risk of disintermediation,” i.e., the possibility publishers and advertisers would transact through rivals. Disintermediation at any level of the ad tech stack was a serious threat to Google’s entire strategy of being the sole entity with end-to-end control over digital advertising transactions.

150. In other documents evaluating whether to buy a yield manager like AdMeld, Google candidly acknowledged that the underlying “technology is irrelevant to us.” Google already had in place the only yield management tool it wanted publishers to use: dynamic allocation’s real-time bidding integration with Google’s ad exchange. Google also recognized that its customer base already largely overlapped, “so we aren’t buying customers.” Only one real question remained open for Google: “How does the competitive landscape change if we buy one?”

151. The Antitrust Division investigated the AdMeld deal before it closed. Like the FTC considering the DoubleClick acquisition, the Antitrust Division declined to challenge the deal based on assumptions about the ad tech market that, with the benefit of hindsight, were incorrect—in no small part due to Google’s subsequent anticompetitive conduct. At the time, the Antitrust Division cited multi-homing among display advertising platforms as a factor that “lessens the risk that the market will tip to a single dominant platform.” But Google’s increasing scale and dominance across the ad tech stack, coupled with its subsequent exclusionary conduct, destroyed the ability of advertisers and publishers to effectively multi-home among alternative ad exchanges. As a result, the market tipped and AdX became the dominant ad exchange.

152. Shortly after the AdMeld deal closed, Google combined the yield management functionality of AdMeld into DFP and migrated all AdMeld customers to AdX. Critically, it then shut down AdMeld’s nascent real-time bidding technology, quashing a competitive threat that otherwise might have challenged Google’s market position and forced Google to move toward a more open system that allowed publishers to utilize AdMeld’s innovative technology to facilitate real-time competition among non-Google ad exchanges and advertisers.

153. By acquiring AdMeld, Google eliminated the existing competition between AdMeld and Google's sell-side products, foreclosed any potential competition, and helped eliminate the leading yield management technology that Google knew might displace its dominant market positions.

2. *Google Doubles Down on Preventing Rival Publisher Ad Servers from Accessing AdX and Google Ads' Demand*

154. After acquiring and killing AdMeld's innovative technology in order to prevent publishers from having the opportunity to experience real-time competition between Google and rival ad exchanges and publisher ad servers, Google clamped down on similar attempts by publishers to allow Google's ad exchange to integrate with rival publisher ad servers.

155. By 2015, Google's publisher ad server, DFP, had reached a 90% market share and had snuffed out most meaningful competition. In part because of the scale that Google's publisher ad server had achieved by excluding competitors, Google's ad exchange was large and growing quickly; Google Ads likewise remained the dominant advertiser ad network and an especially valuable source of advertising demand for many publishers. Emboldened by its success, in 2014 Google changed the AdX terms of service to further entrench its market power. Those changes prohibited publishers from using non-Google ad servers, or the remaining yield management solutions, to compare bids from Google's ad exchange with bids from other ad exchanges in real time, notwithstanding the increased access to inventory such an integration could provide to advertisers buying on AdX. In effect, Google decreed that any publisher that wanted real-time competition involving AdX would have to use Google's publisher ad server,

DFP, formally cementing in policy what Google had intended from the outset of its relaunch of AdX in 2009.

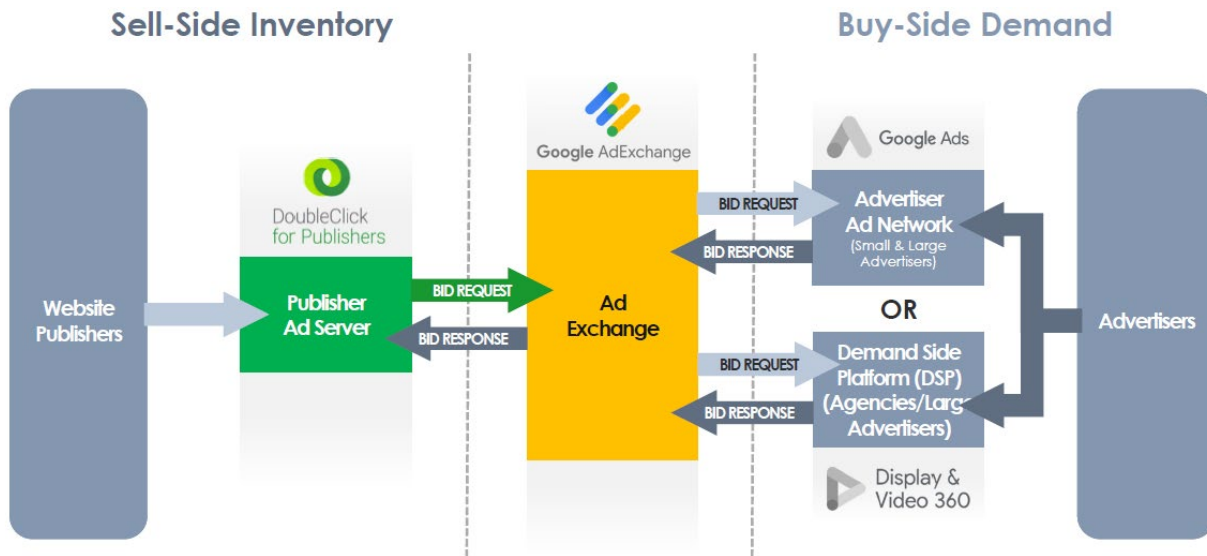


Fig. 13

156. Google's decision was bad for publishers, locking them into a less innovative publisher ad server with artificial limitations on real-time price competition for advertising inventory. It was also bad for any would-be publisher ad server rivals—effectively sounding the death knell for future publisher ad server competition. Google's exclusionary policy effectively prohibited a competing publisher ad server from offering any form of real-time competition that included Google's ad exchange and the unique advertiser demand that came with it. Forgoing such competition was a non-starter for nearly all publishers. This restriction is still in place today, an insurmountable obstacle for any nascent publisher ad server competitor.

157. Google built a wall around its exclusive link between its publisher ad server and ad exchange because it feared competition. In particular, Google feared a rival could offer a more attractive publisher ad server by simply allowing all advertiser demand to compete in real time on a level playing field for publisher inventory. More demand competing in real time for

publisher inventory generally increases the likelihood that the advertiser that is willing to pay the most for an impression will have a chance to buy it. Rivals that offered technology upending this policy would be seen as offering a better publisher ad server. As one Google employee wrote, if another publisher ad server could place Google's ad exchange in real-time competition with other ad exchanges, that ad server could offer publishers a "super set of demand" and "[n]o one would sign up for AdX directly" through Google's publisher ad server.

158. Even though both publishers and advertisers benefit from real-time competition between AdX and other ad exchanges, by policy, Google limited real-time competition from rival ad exchanges to maintain its dominant positions at both ends of the ad tech stack and to further insulate its growing position in the ad exchange market. Google's decision was based on business, not technology. As the lead architect of AdX explained in an internal email about the policy, "Our goal should be all or nothing – use AdX as your SSP [ad exchange] or don't get access to our demand." Indeed, Google had already worked quietly to develop the technology that might allow AdX to integrate in real time with non-Google publisher ad servers. But Google made a "strategic decision" to prohibit such integrations via contract; it terminated its internal projects and blocked efforts by rivals and publisher customers to implement such integrations. That prohibition endures today, and both publishers and advertisers are paying the price for Google's anticompetitive refusal to innovate or integrate.

159. Now, the only way a publisher can access Google's ad exchange outside Google's publisher ad server is by placing an "AdX Direct" tag on the publisher's website. Even though these tags could benefit buyers on Google's ad exchange by providing access to additional publisher inventory, Google designed the tags to discourage publishers from using them. They offer only the most rudimentary functionality: publishers can send a request to Google's ad

exchange with a price floor, and if there is an advertiser on AdX willing to pay that price or higher, Google's ad exchange wins the inventory. No other competing bids are considered, and Google's bid cannot be compared to other ad exchanges' bids.

160. Recognizing that AdX Direct is an antiquated relic in comparison to real-time bidding, Google even planned to eliminate the tag entirely in 2019. Google later paused that project as antitrust enforcers focused their gaze on the company's digital advertising business. But Google has not retained AdX Direct because it is a competitive product offering valued by publishers. Rather, in the words of a Google employee, it merely serves as "a concept for antitrust"—something Google's antitrust lawyers could claim offers rival ad servers some remote chance of competing on the merits with Google's ad server. Google's internal analyses of AdX Direct, however, reflect publishers' reality: Google's restrictions make impossible any reasonable substitute for the real-time integration with Google's ad exchange available exclusively through Google's ad server.

3. *Google Manipulates Google Ads' Bidding Strategy to Block Publisher Partnerships with Rivals*

161. Google also took the opportunity to tweak its Project Bernanke algorithm to further lock in publishers who considered using innovative bidding technology offered by Google's rivals. Some publishers attempted to partner with rival ad exchanges to offer them "first look" access to inventory—an opportunity to bid in real time for inventory before it was offered to Google's AdX. "First look" could potentially prop up rival ad exchanges by giving them effective access to some of the most valuable inventory.¹⁷

¹⁷ Previously, Google used dynamic allocation in its publisher ad server to exclude rival exchanges from meaningful competition. By only permitting Google's own ad exchange to bid in real time—ahead of any other exchange—for impressions, Google was unfairly advantaged and competitors were effectively stymied from competing.

162. Beginning in 2014, Google recalibrated the earlier Project Bernanke to decrease Google Ads' bids on AdX for publishers that allowed rivals an opportunity to buy inventory ahead of AdX. Google called this "Project Bell." Where publishers tried to partner with rival sources of advertising demand for "first look" access to inventory, Google reduced bids—without any input from or awareness of the underlying advertiser—by about 20%. According to Google's documents, it explicitly warned publishers that utilizing innovative "first-call" technology from rivals would cause publisher yield to drop 20 to 30%. Of course, only a company like Google with substantial market power across the entire ad tech stack would have the incentive or ability to implement such a program. Project Bell both insulated Google's ad exchange from this new form of competition and preserved preferential access for buyers on Google's ad exchange, including Google Ads.

D. Google Responds to the Threat of Header Bidding by Further Excluding Rivals and Reinforcing Its Dominance

1. The Industry Attempts to Rebel Against Google's Exclusionary Practices

163. By 2015, Google's publisher customers and ad exchange competitors had grown so frustrated with Google that they attempted to implement a form of open, real-time competition with Google's ad exchange that evaded Google's exclusionary restrictions. This innovative technology was called header bidding.

164. Header bidding worked as follows: publishers inserted certain computer code into the "header" section of the HTML code of a web page. This code triggered a real-time auction among ad exchanges *before* the publisher's web page called the publisher ad server.¹⁸ The highest bid from the header bidding auction was then sent to the publisher's ad server. Because

¹⁸ Because early versions of the header code were run on the device of the user, or client, they were referred to as "client-side" header bidding.

of the way Google configured DFP, the winning bid from the header bidding auction was then sent to Google's ad exchange to see if it could beat that price. Critically, through dynamic allocation, Google's ad exchange always received this "last look" advantage, essentially a right to buy any impression as long as it had at least one advertiser willing to match the competing bid price from the header bidding auction.

165. As originally designed, header bidding had limitations that stemmed from Google's restrictions on how publishers could sell their inventory through Google's publisher ad server. For example, publishers had to configure thousands of entries into the publisher ad server and recode their pages to implement a workaround to enable header bidding. Web pages also ran somewhat more slowly because publishers had to run multiple auctions sequentially: the header bidder auctions first, and then Google's ad exchange auction, which always ran last. Despite these limitations, for the first time, Google's ad exchange was forced to compete, at least in some fashion, against real-time bids from rival ad exchanges rather than against static, historical average prices from those ad exchanges. In assessing the impact of header bidding, a 2016 Google internal presentation noted "header bidding and header wrappers are BETTER than [Google's platforms] for buyers and sellers." Google explained that competition between AdX and buyers using header bidding increased publisher revenues by 30 to 40%, and provided additional transparency to advertisers. In essence, header bidding allowed publishers, advertisers, and Google's rivals an opportunity to at least partially circumvent Google's restrictions against real-time competition. Market participants had demonstrated their preference for improved choice, flexibility, and competition, even if it came at the cost of burdensome computer workarounds and slower load times for end users.

How Does Header Bidding Work?

Header bidding allows pubs to get a real-time price signal before allocating an impression to their ad server

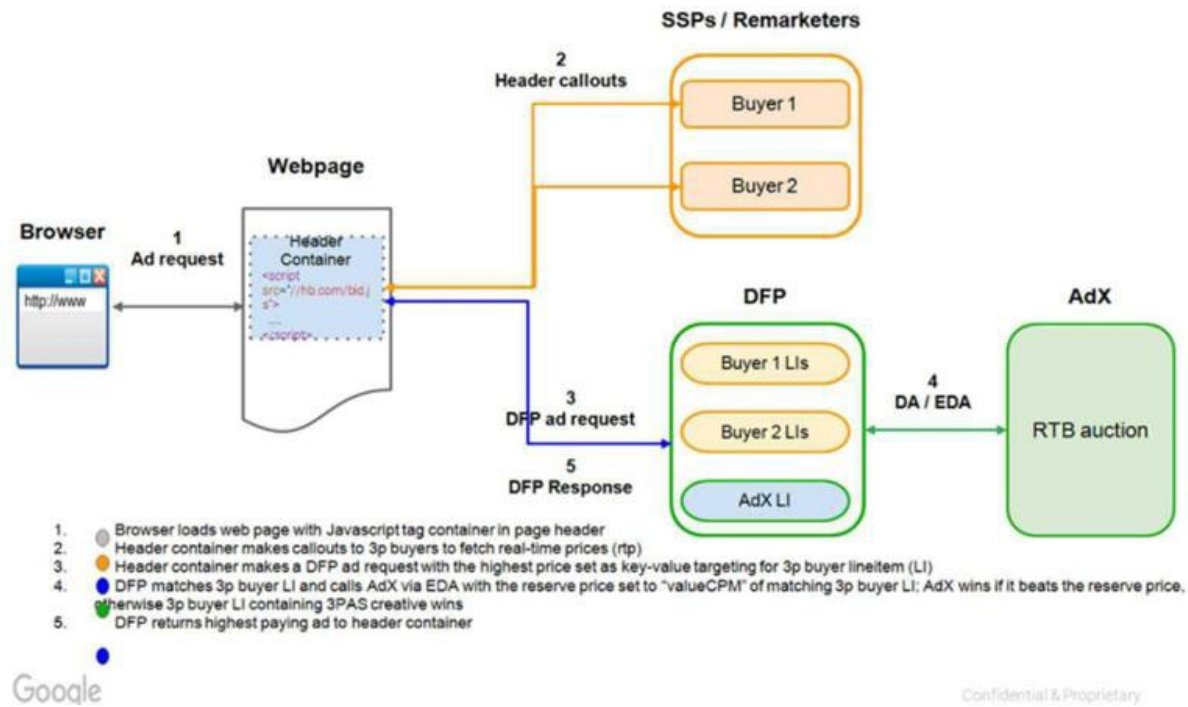


Fig. 14¹⁹

166. In practice, header bidding dramatically improved the competitiveness of rival ad exchanges. Header bidding provided a real opportunity for rival ad exchanges to see and compete for more publisher inventory, and potentially gain scale to compete effectively with Google. By allowing a publisher to call multiple ad exchanges in real time—effectively multi-homing at the ad exchange level—header bidding vastly increased the amount of inventory rival ad exchanges could offer their advertisers. In turn, advertisers had the opportunity to see and bid on more inventory—potentially through lower-cost channels than Google’s ad tech tools—increasing their chances of winning inventory. By improving the ability of advertisers and publishers to connect, these rival ad exchanges were able to clear more transactions, increase

¹⁹ “SSPs” refers to non-Google ad exchanges.

revenues for publishers, and improve the quality of matches. In turn, header bidding had the potential to attract more advertisers and publishers to these rival ad exchanges by increasing the incremental value they could offer.

167. Due to this increased competitiveness, non-Google affiliated buyers began to buy more advertising inventory through third-party ad exchanges using header bidding. Google's internal analysis showed a deceleration in spend by non-Google advertiser buying tools on AdX as "header bidding removed AdX inventory exclusivity . . . [and] buyers shift[ed] spend as other inventory sources delivered equal/better value." A large buyer explicitly indicated to Google that they were "shifting spend over to HB partners" because they were "seeing better performance."

168. More transactions flowing through rival ad exchanges made it easier for those ad exchanges to offset the massive cost of processing billions of ad requests each day; ad exchanges are only compensated for requests that result in a won transaction. These new transactions also provided ad exchanges with additional data on the universe of publisher inventory, user targeting data, and the competitive landscape.

169. Building on the success of early client-side header bidding, several companies invested to develop new innovative free or low-cost tools (called "wrappers") that enabled "server-side" header bidding. This new form of header bidding allowed the header of a web page to call a single server, which then sent calls to multiple ad exchanges, each of which returned a bid to the server, which in turn passed on the winning bid to the web page. Server-side header bidding turbocharged the scale benefits of header bidding by decreasing integration costs and improved the internet user experience by reducing latency introduced by header bidding.

170. Internally, Google recognized that header bidding substantially benefited every market participant except one: Google. For that reason, Google refused to participate and instead

chose to stifle any competitor that dared employ header bidding. As one Google employee explained, “[Header bidding] gives many publishers better yield, so it’s a no-brainer for a publisher to adopt it.” A late 2015 internal discussion somberly noted that Google “[did] not have incredibly robust arguments to discourage header bidding” and conceded that header bidding offered the competition Google had publicly preached but privately precluded:

With AdX we’ve always advocated the more competition a pub has being considered with real time price competition the better the yield. Our competition is using this same argument for why header bidding makes sense. If they can submit a near real time price into DFP the[] competition with AdX is improved.

As another Google employee observed, “[Google’s ad server] has historically made it difficult for [ad exchanges] to compete on a level playing field with AdX.”

171. Google viewed header bidding—and particularly server-side header bidding—as a direct and, in the words of a 2016 internal strategy paper, “existential threat” to the market power Google had amassed. Internal Google documents confirm that Google understood header bidding to be a direct response by its customers and competitors to counteract Google’s increasing dominance and its “unwillingness to open our systems to the types of transactions, policies and innovations that buyers and sellers wish to transact.” Header bidding was an attempt to “circumvent dynamic allocation,” one Google employee noted in late 2015. Another employee recognized that “[p]ublishers felt locked-in by dynamic allocation in [Google’s ad server] which only gave [Google’s ad exchange the] ability to compete, so HB was born.” Another described header bidding as a “world of true, multi-sourced [real-time bidding]” without Google as the “authoritarian intermediary.”

172. Beyond breaking the restrictions Google had put in place, header bidding also represented a pervasive threat to Google’s market power stemming from its unique and